

Electronics & Physical Computing (SCS2315/IS2207)

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Course Outline

- Fundamentals – Ms. Nadeesha
- Circuits & Components – Dr. Hiran
- Digital Electronics – Ms. Nadeesha
- Physical Computing & IoT – Dr. Hiran

Assessment

- **Continuous Assessment – 40%**
 - In-class assignments
 - Activities
 - Practical test
- **Final Written Paper – 60%**
 - MCQs and structured questions

Electronic Components

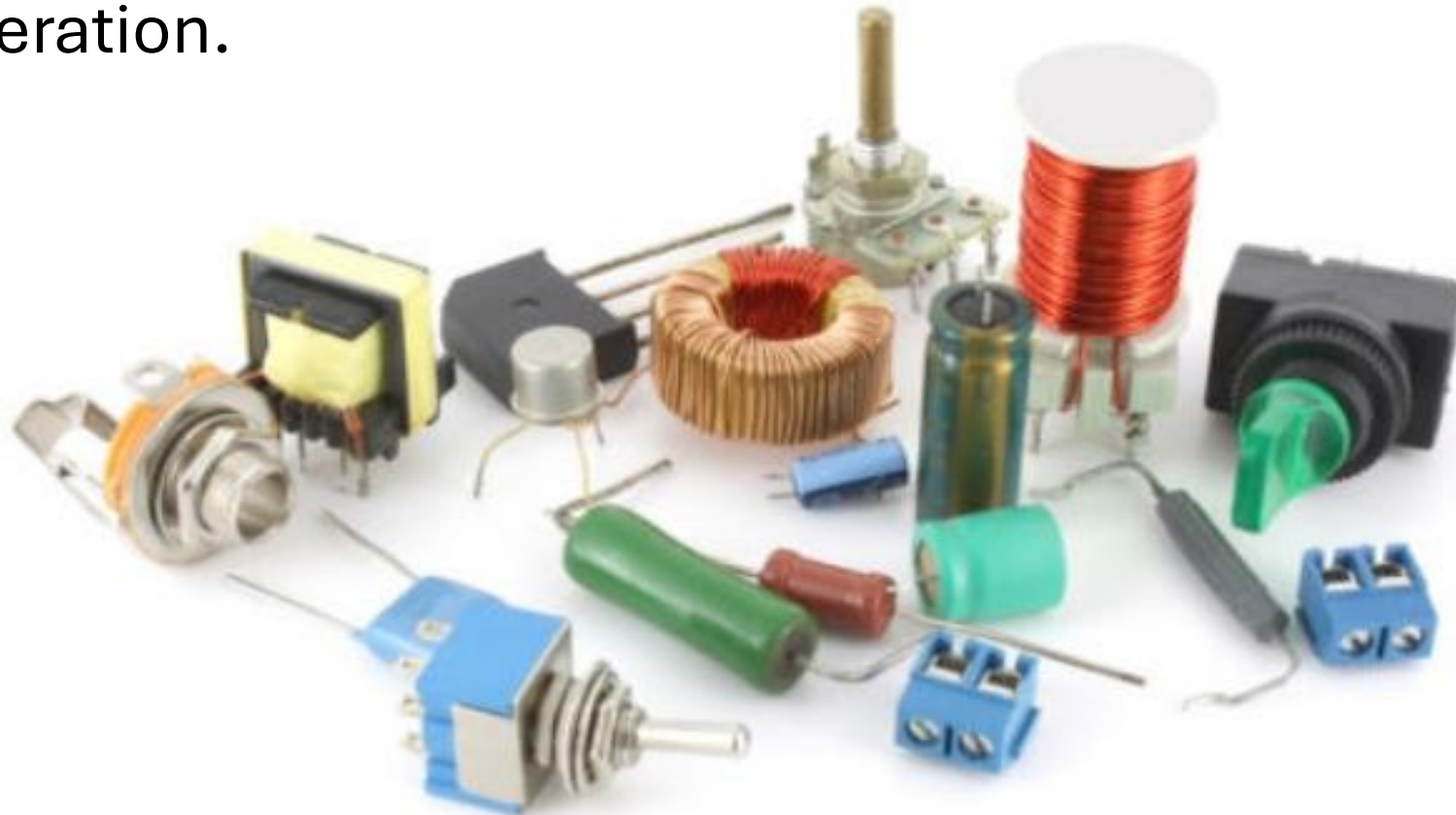
Dr. Hiran Ekanayake

Electronic Components

- Passive Components
- Active Components
- Electromechanical Components
- Semiconductor Devices
- Power Components
- Sensors & Transducers
- Communication Components
- Protection Components
- Display & Output Devices
- Embedded & Digital Components
- Interconnection Components
- Etc.

Passive Components

- Passive components do not amplify signals or require external power for operation.
 - Examples:
 - Resistor
 - Capacitor
 - Inductor
 - ...
- 
- A photograph showing a variety of passive electronic components arranged on a white surface. The components include several resistors of different shapes and sizes, a large electrolytic capacitor with a blue and silver body, a toroidal inductor with orange wire, a solenoid with red wire, a transformer with a yellow core, and a green potentiometer.



Resistor (R)



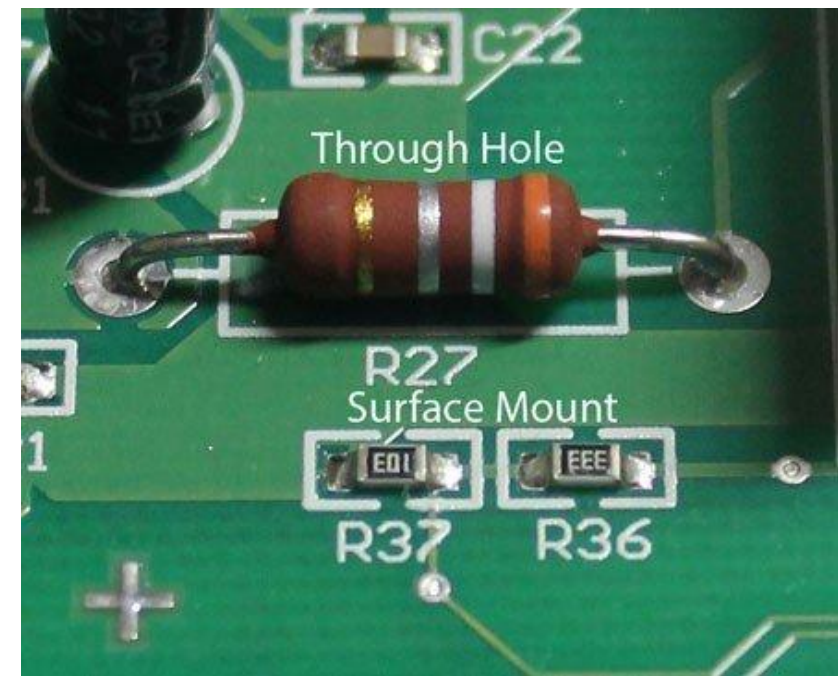
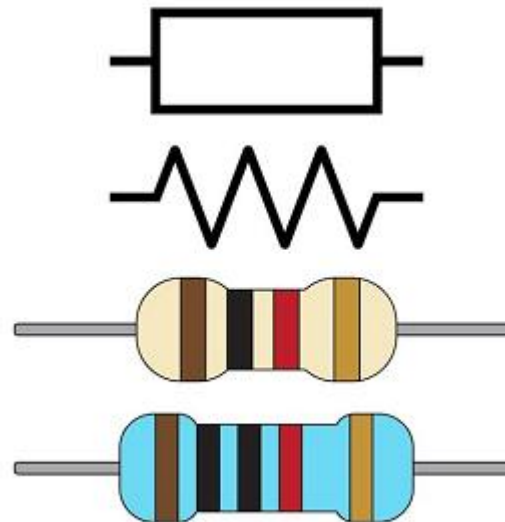
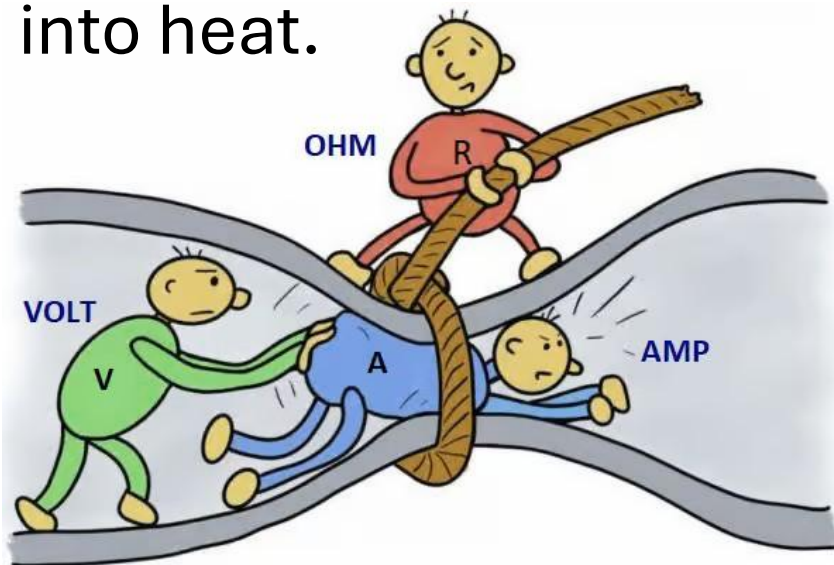
Online Tutorial on Resistor:

https://www.electronics-tutorials.ws/resistor/res_1.html

- ▶ 1. Types of Resistor
- 2. Resistor Colour Code
- 3. Resistors in Series
- 4. Resistors in Parallel
- 5. Resistors in Series and Parallel
- 6. Potential Difference
- 7. Resistor Power Rating
- 8. Resistors in AC Circuits
- 9. Resistor Tutorial Summary
- 10. Varistor Tutorial
- 11. Resistor Colour Code Wheel
- 12. Potentiometers
- 13. Resistivity
- 14. LED Resistor

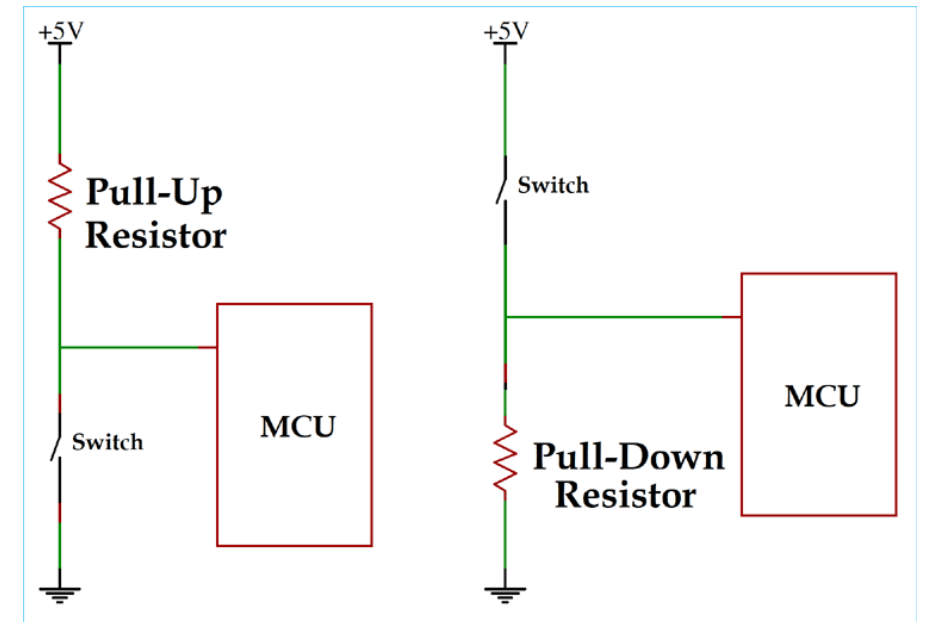
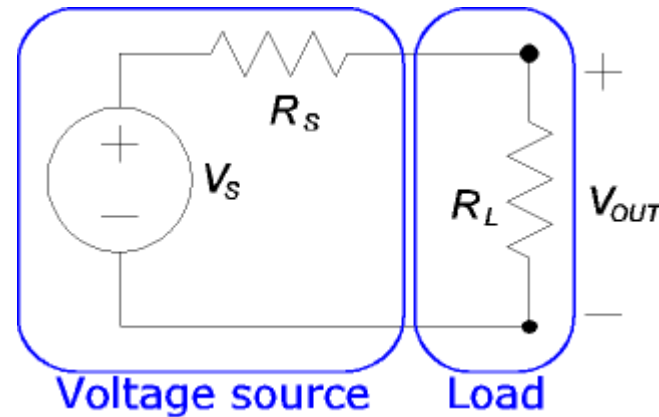
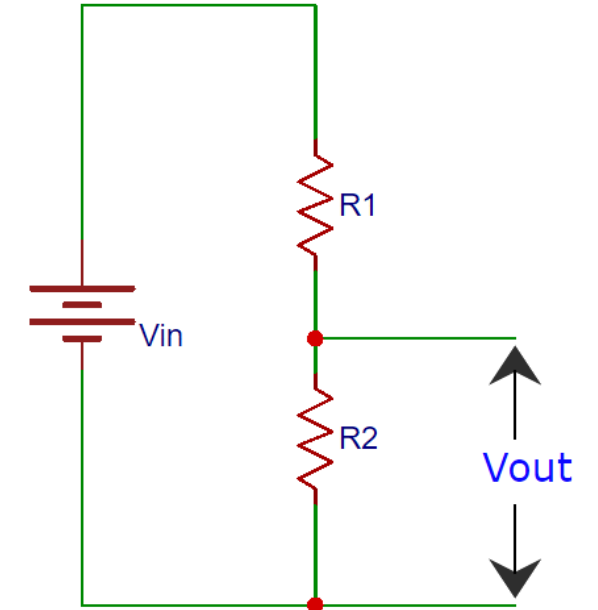
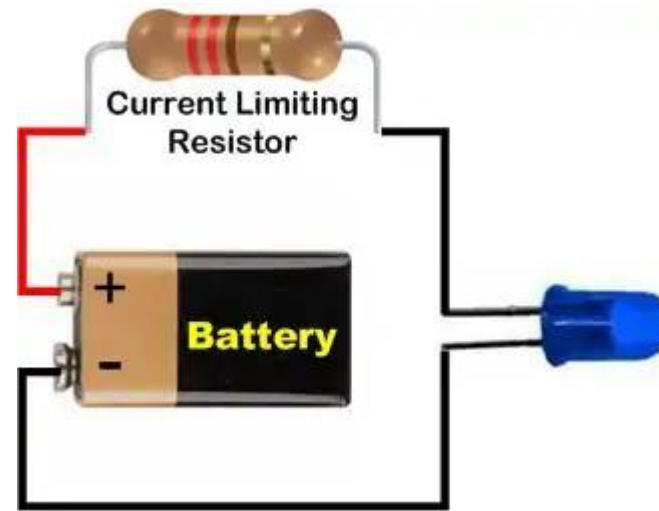
What is a Resistor?

- A passive component that **opposes or limits the flow of electric current** in a circuit.
- It does this by converting electrical energy into heat.

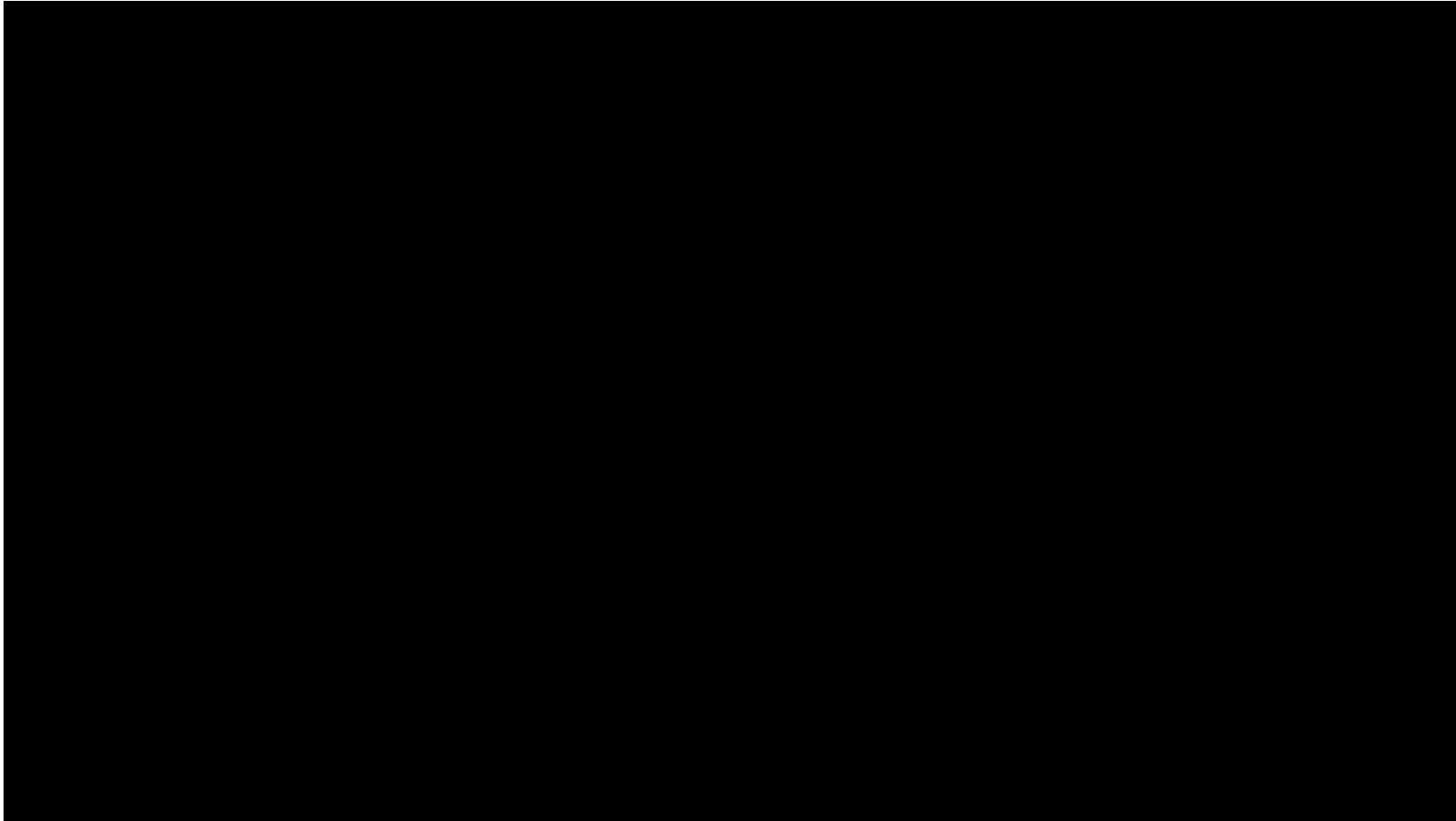


Purpose of Resistor

- Limit current
- Divide voltage
- Bias transistors
- Pull-up / pull-down
- Load resistance
- Etc.



Resistor Current Limiting

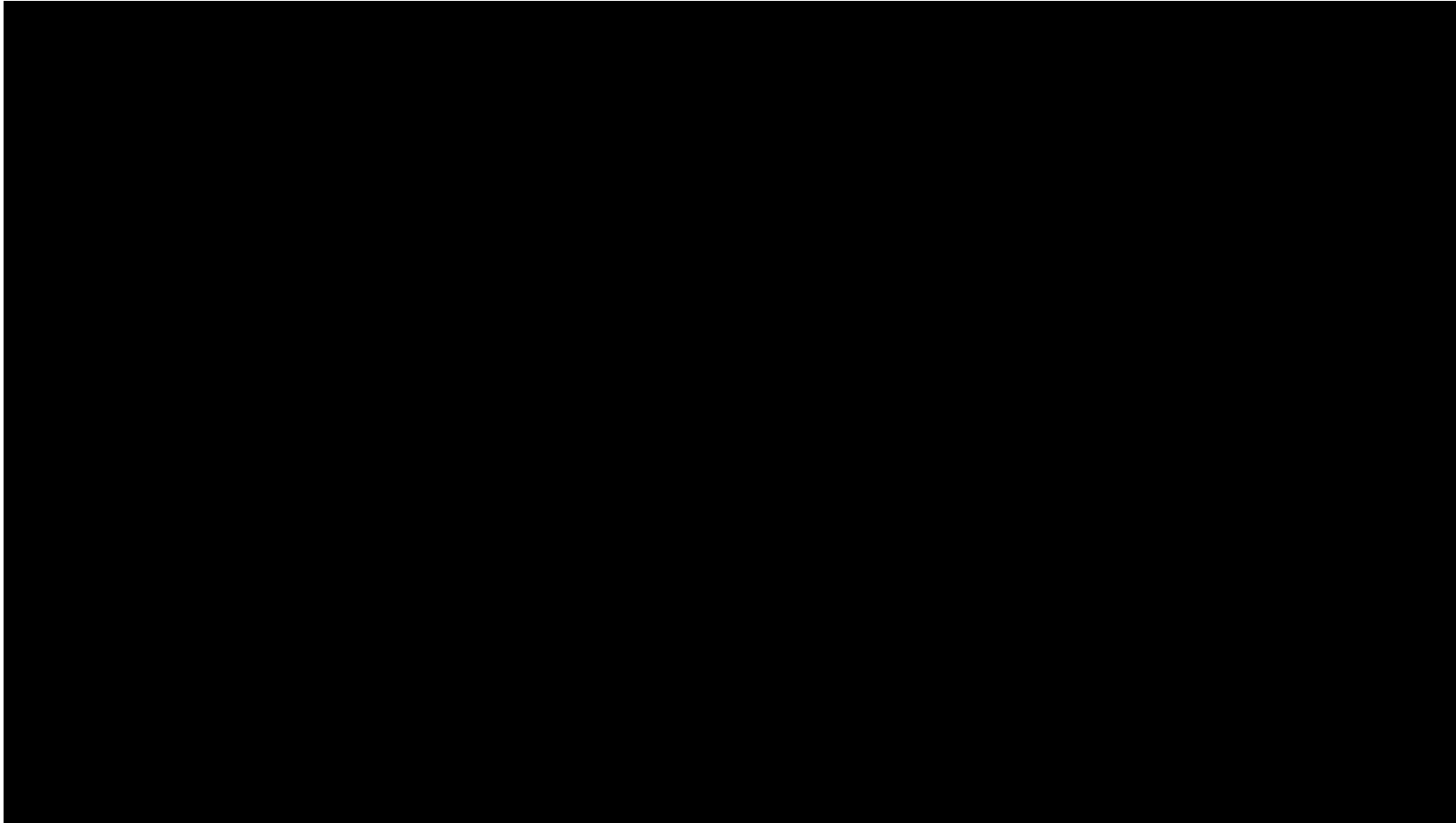


Unit of Resistance

- The SI unit of electrical resistance is **Ohm (Ω)**.
- Prefixes:

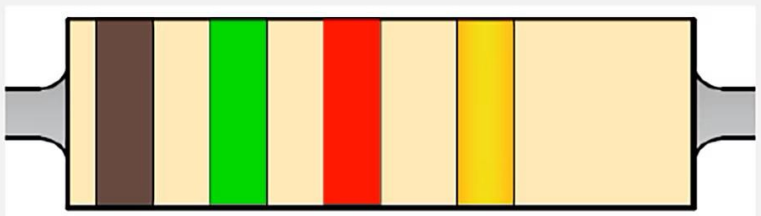
Unit	Symbol	Value
Milliohm	m Ω	0.001 Ω
Ohm	Ω	1 Ω
Kilo-ohm	k Ω	1,000 Ω
Mega-ohm	M Ω	1,000,000 Ω

Measure Resistance Using a Multimeter Demo



Resistor Color Code

- Instead of printing numbers on small resistors, colored bands are used.
- 4-Band Resistor:
 - Band 1 → First digit
 - Band 2 → Second digit
 - Band 3 → Multiplier
 - Band 4 → Tolerance
- E.g.: Brown Black Red Gold →
 $1\ 0 \times 100 = 1000\Omega = 1\text{k}\Omega \pm 5\%$



	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	x 1	
Brown	1	1	x10	±1%
Red	2	2	x10 ²	±2%
Orange	3	3	x10 ³	±3%
Yellow	4	4	x10 ⁴	±4%
Green	5	5	x10 ⁵	±0.5%
Blue	6	6	x10 ⁶	±0.25%
Violet	7	7	x10 ⁷	±0.1%
Grey	8	8	x10 ⁸	±0.05%
White	9	9	x10 ⁹	
Gold			x10 ⁻¹	±5%
Silver			x10 ⁻²	±10%

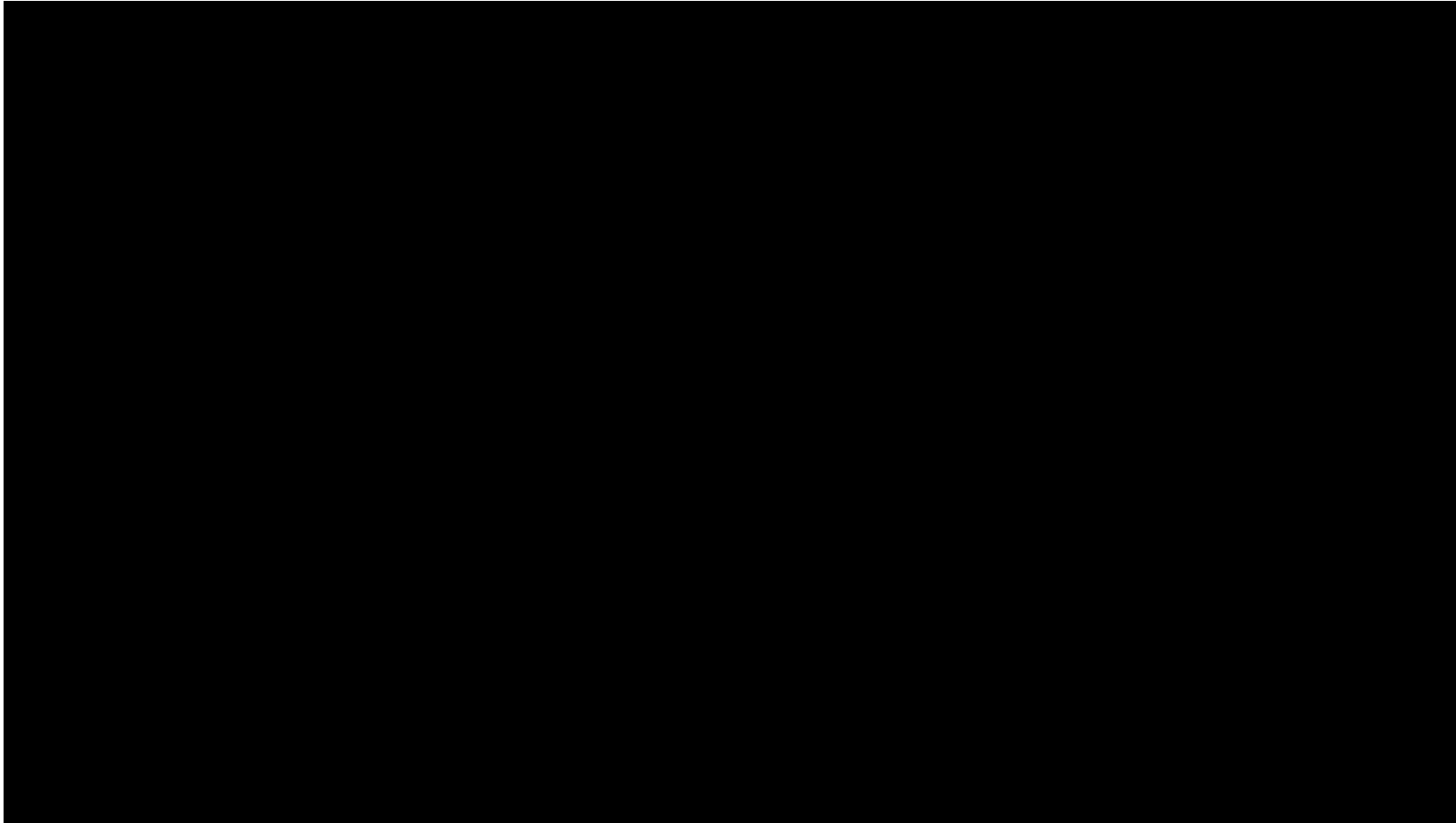
Ohms Law

- Ohm's Law describes the relationship between **Voltage (V)**, **Current (I)**, and **Resistance (R)** as:

$$V = IR$$

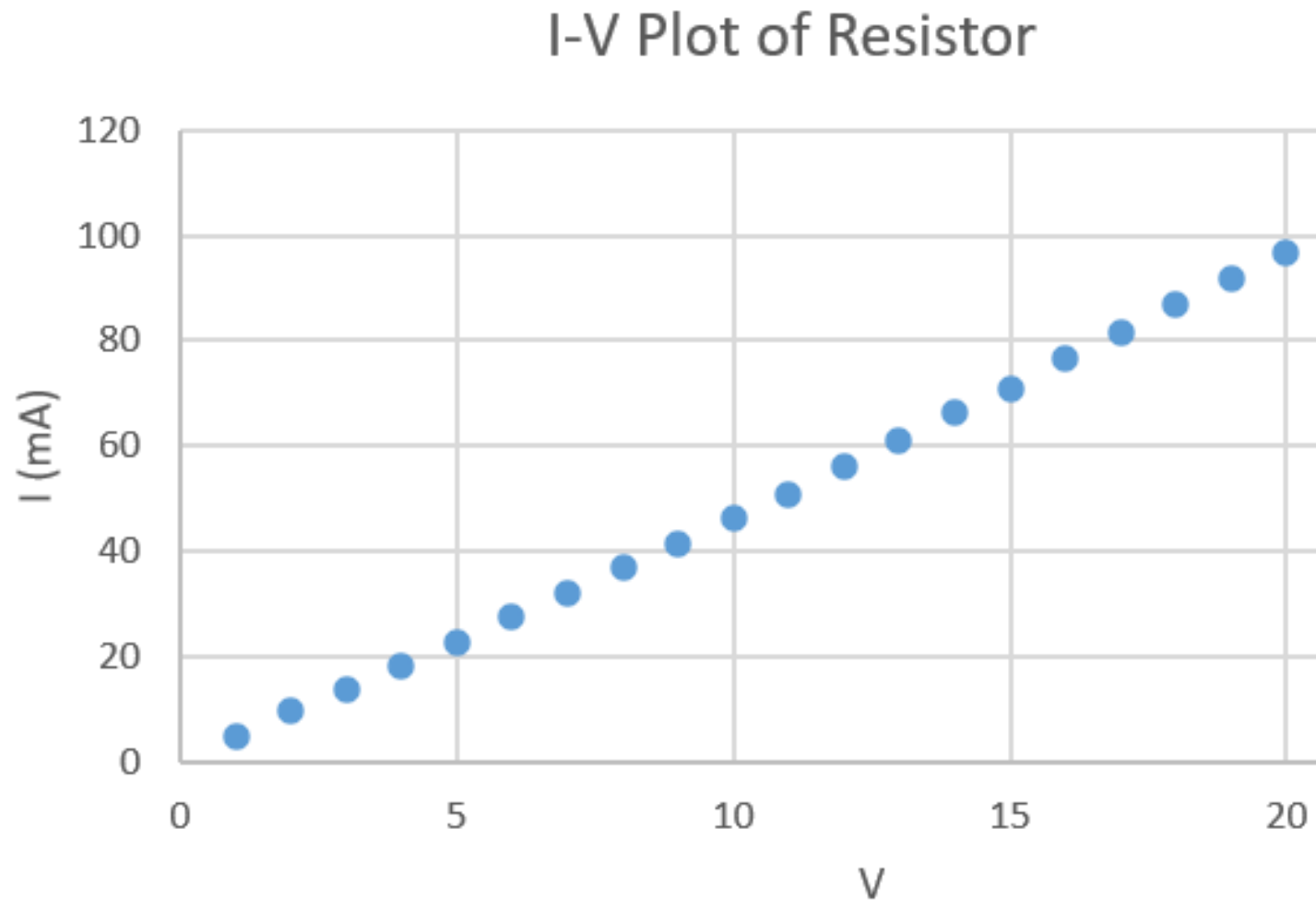
- Exercise:
 - (1) A resistor of **470Ω** is connected to a **12V** supply. Find the current.
 - (2) A circuit connected to **9V** draws **0.03A**. Find the resistance.
 - (3) A current of **2A** flows through a **6Ω** resistor. Find the voltage across the resistor.
 - Answers: (1) 25.5 mA, (2) 300Ω, (3) 12V

Measurement of the I–V Relationship of a Resistor

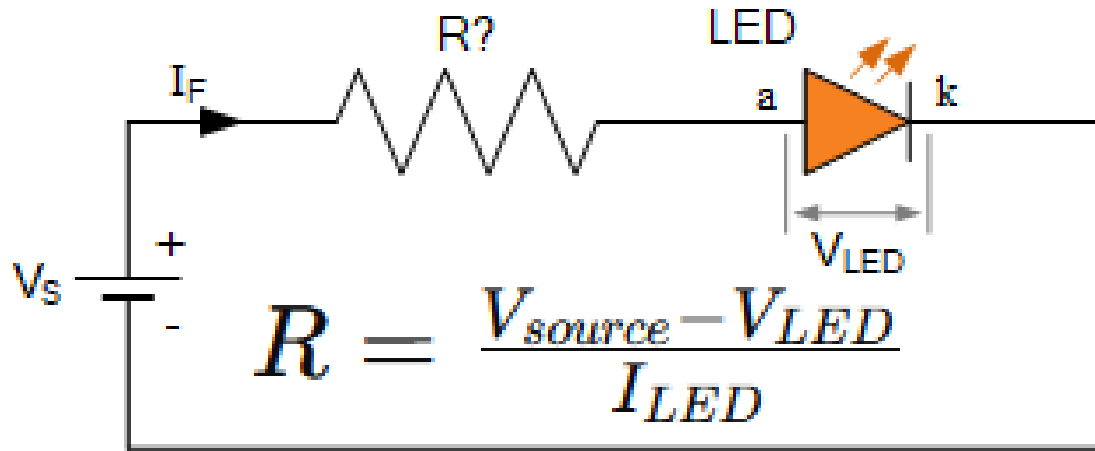


Measurement of the I–V Relationship of a Resistor

V	I (mA)	P=VI	R=V/I
1	4.93	0.00	203
2	9.47	0.02	211
3	13.8	0.04	217
4	18.28	0.07	219
5	22.8	0.11	219
6	27.3	0.16	220
7	32	0.22	219
8	36.9	0.30	217
9	41.2	0.37	218
10	46.1	0.46	217
11	50.8	0.56	217
12	56.1	0.67	214
13	60.9	0.79	213
14	66.2	0.93	211
15	71	1.07	211
16	76.6	1.23	209
17	81.6	1.39	208
18	86.8	1.56	207
19	91.6	1.74	207
20	96.8	1.94	207



LED Series Resistor Calculation



- R = resistor value (Ω)
- V_{source} = supply voltage
- V_{LED} = LED forward voltage
- I_{LED} = desired LED current

E.g., Given that supply = 9V,
white LED = 3.2V, and current
= 15mA $\rightarrow R \approx 387\Omega = \mathbf{390\Omega}$

LED Color	Typical Forward Voltage
Red	1.8V – 2.2V
Yellow	2.0V – 2.2V
Green	2.0V – 3.2V
Blue	3.0V – 3.6V
White	3.0V – 3.6V

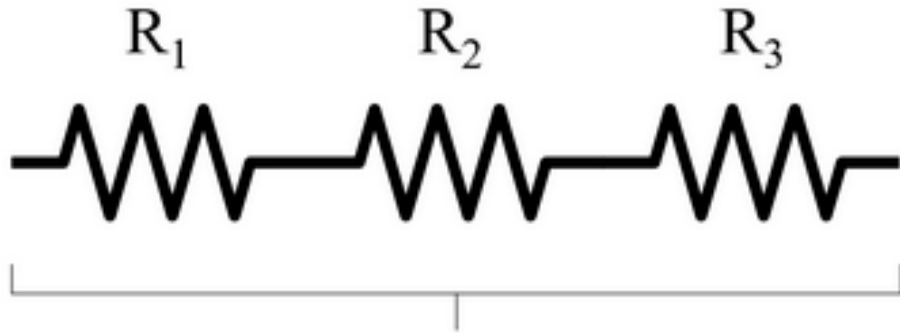
Common indicator LEDs:

- 5 mA $\rightarrow$ low brightness
- 10 mA $\rightarrow$ normal brightness
- 20 mA $\rightarrow$ maximum common safe current

Resistor Configurations

Series Resistors

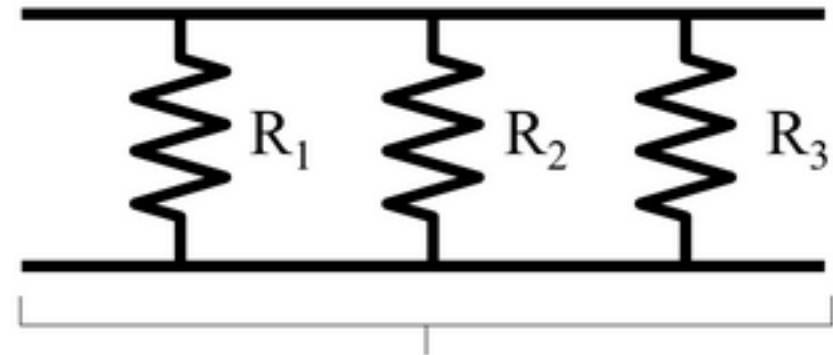
Current is the same across each resistor. Voltage is divided.



$$R_{Equivalent} = R_1 + R_2 + \dots + R_{N-1} + R_N$$

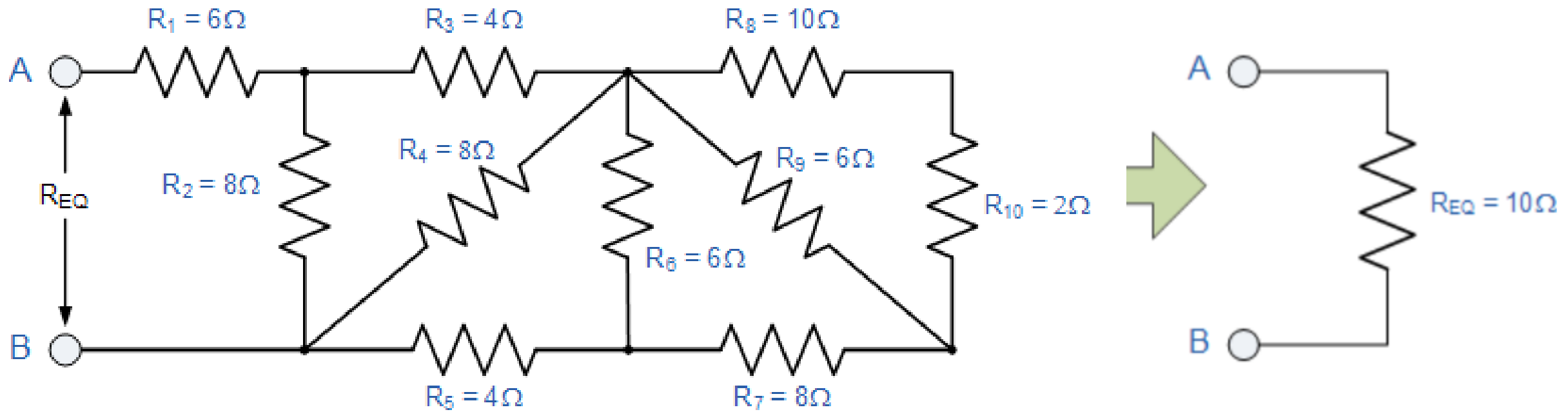
Parallel Resistors

Voltage is the same across each resistor. Current is divided.

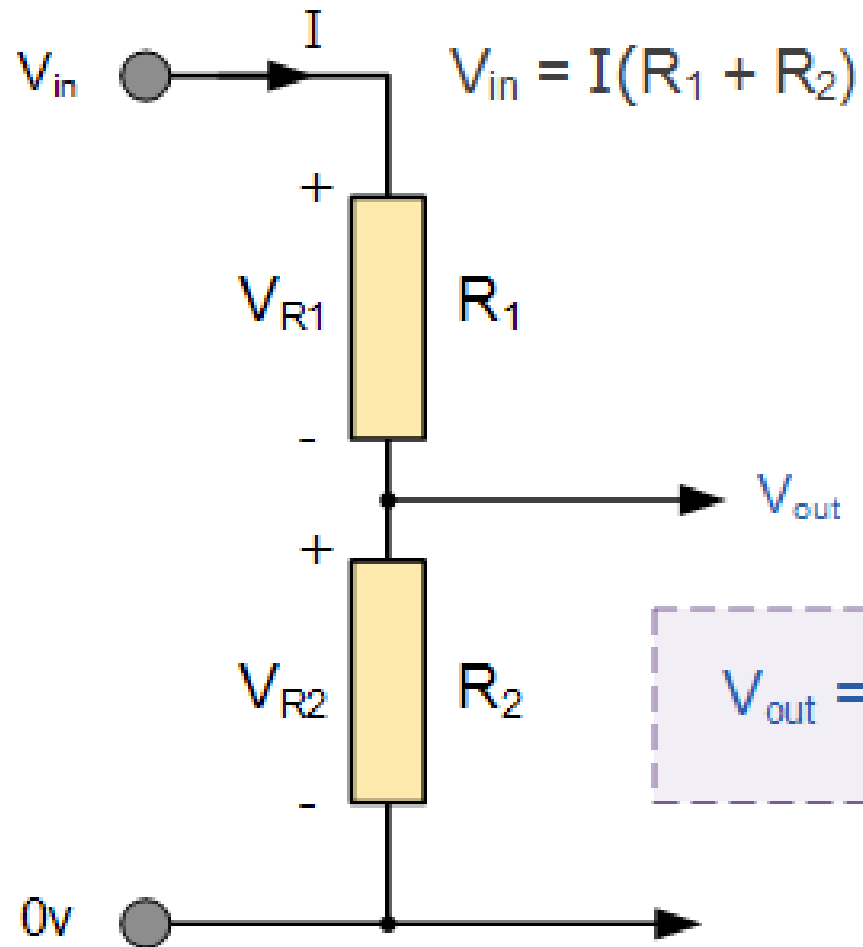


$$R_{Equivalent} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_{N-1}} + \frac{1}{R_N}}$$

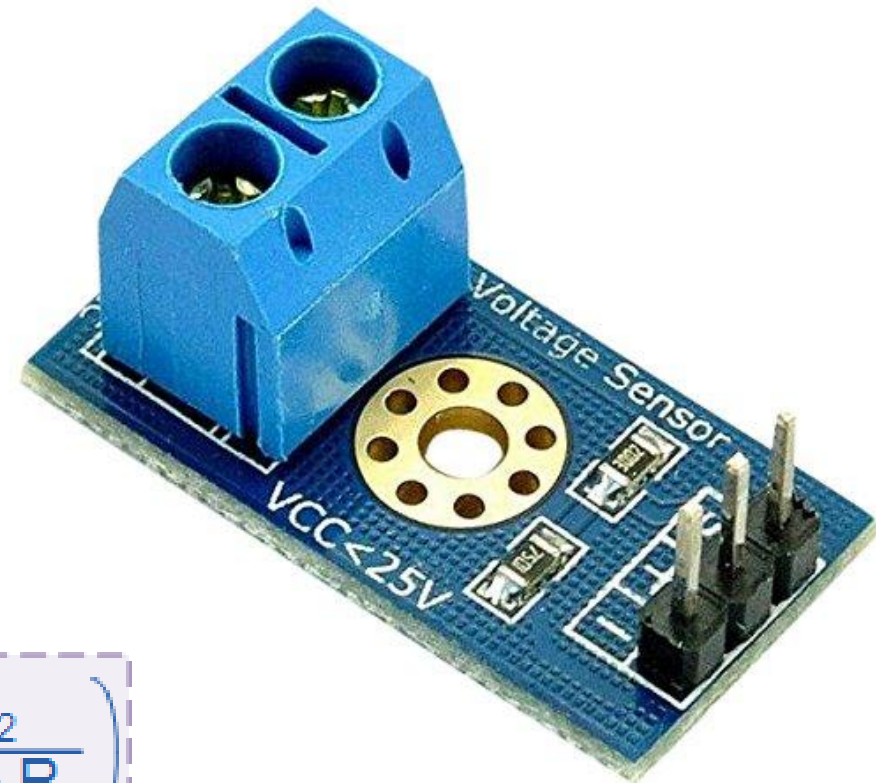
Exercise



Voltage Divider Network



$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

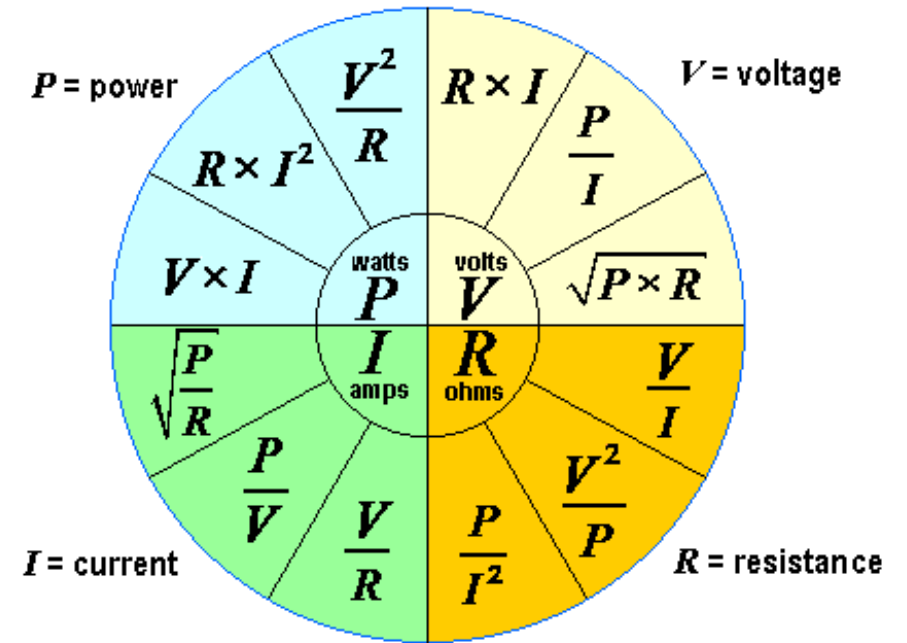


Power Dissipation

- Power dissipation is the **actual electrical power converted into heat** inside the resistor during operation, which is given by the formulas:

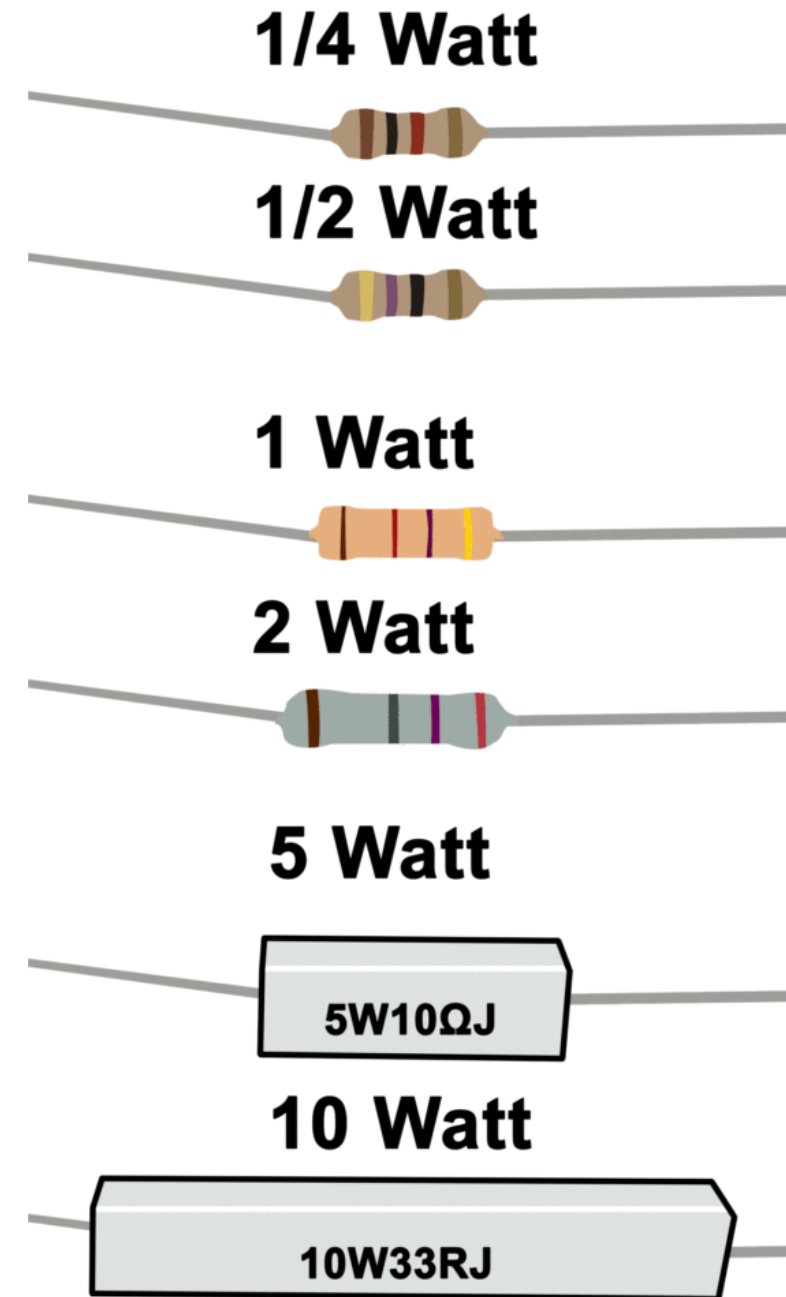
$$P = VI = I^2R = \frac{V^2}{R}$$

- Where,
 - P = dissipated power (Watts)
 - V = voltage across resistor
 - I = current through resistor
 - R = resistance



Power Rating

- Power rating is the **maximum power the resistor can safely dissipate continuously** without overheating or being damaged.
- Typical ratings: 1/8 W, 1/4 W, 1/2 W, 1 W, 5 W, and 10 W
- This rating is determined by the physical size, material, cooling ability, and construction of the resistor.



The Connection Between Power Dissipation and Power Rating

- The actual dissipated power must always remain below the resistor's power rating, i.e., $\text{Power Dissipation} \leq \text{Power Rating}$.
- Common practice is to use only 50%–60% of rated power.
 - E.g., If resistor dissipates 0.2 W, then use at least a 0.5 W resistor.



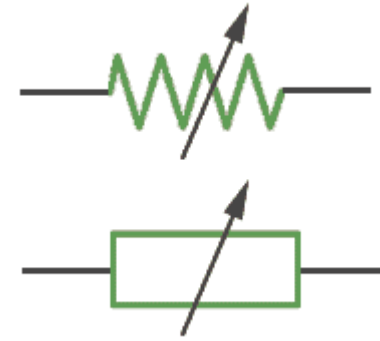
Resistor Tolerance

- Tolerance tells how much the actual resistance may vary from the stated value.
 - E.g., the actual resistor value of a **1k Ω $\pm$ 5%** can lie between 950 Ω (1000 - 5%) to 1050 Ω (1000 + 5%).
- Common tolerance values: $\pm$ 20%, $\pm$ 10%, $\pm$ 5%, $\pm$ 2%, $\pm$ 1%, $\pm$ 0.5%, and $\pm$ 0.1%.
- Tolerance color codes: Gold $\rightarrow$ $\pm$ 5%, Silver $\rightarrow$ $\pm$ 10%, Brown $\rightarrow$ $\pm$ 1%, Red $\rightarrow$ $\pm$ 2%, Green $\rightarrow$ $\pm$ 0.5%, Blue $\rightarrow$ $\pm$ 0.25%, Violet $\rightarrow$ $\pm$ 0.1%, and No band $\rightarrow$ $\pm$ 20%.
- Higher precision resistors cost more.

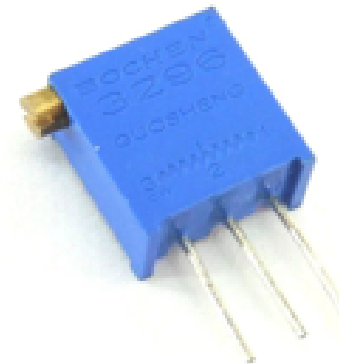
Resistor E-Series

- The E-series defines standard resistor values for each tolerance class.
- The “E” means “**Exponential**” and the values are logarithmically spaced.
- Main E-Series: E6, E12, E24, E48, E96, and E192. The number indicates how many standard values exist per decade.
- A “decade” means between 10Ω to 100Ω , 100Ω to $1k\Omega$, $1k\Omega$ to $10k\Omega$, etc.
- E.g., the E12 series has 12 standard values per decade: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, and 82.

Types of Resistors

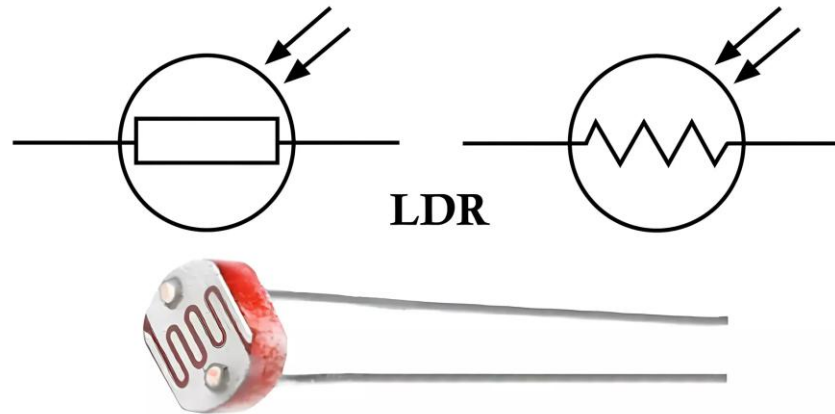


- Fixed Resistors: Value is fixed
 - Carbon Composition, Carbon Film, Metal Film, Metal Oxide Film, Wire Wound, Thick Film, Thin Film, Foil Resistor, Fusible Resistor, Cement Resistor, etc.
- Variable Resistors: Resistance can be adjusted manually
 - Potentiometer, Rheostat, Trimmer Potentiometer, Digital Potentiometer



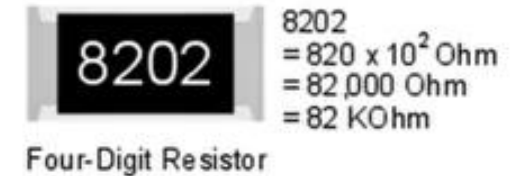
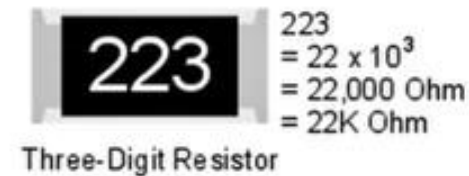
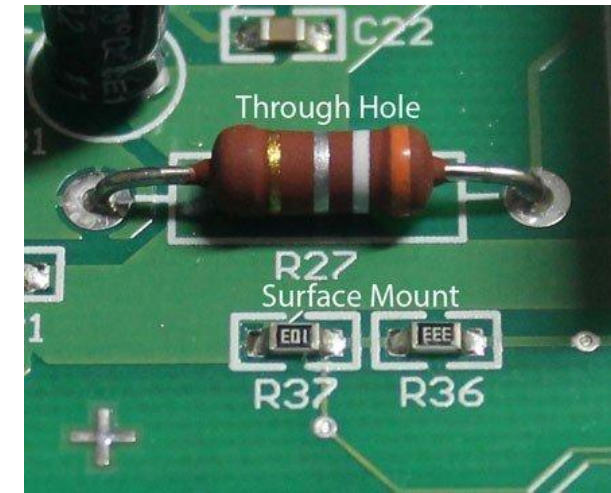
Types of Resistors Cont.

- Dependent / Nonlinear Resistors: Resistance changes according to environmental conditions
 - Thermistor, LDR (Light Dependent Resistor), Varistor, Magneto-resistor, etc.
- Special Purpose Resistors
 - Shunt Resistor, Current Sense Resistor, etc.



Resistor Packaging / Mounting Types

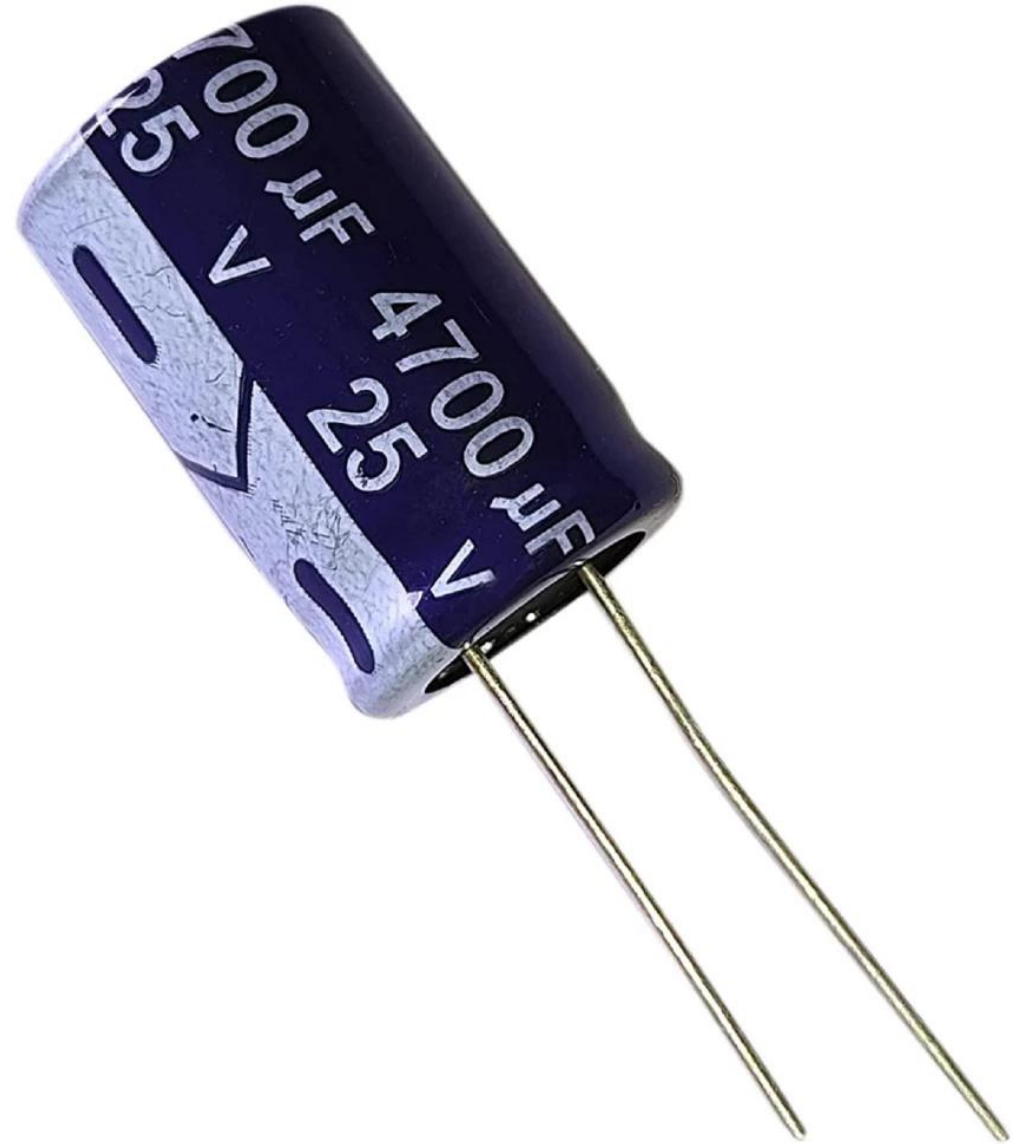
- Through-Hole (THT)
 - Have leads that pass-through holes in the PCB.
- Surface Mount (SMD/SMT)
 - Tiny size and mounted directly on PCB surface.
 - E.g., 0201: 0.02" × 0.01" or 0.6 × 0.3 mm
- Specialized Packages
 - Multiple resistors in one package.



Activity

- Investigate the local availability and prices of 100 Ω and 1 k Ω resistors. Compare prices based on:
 - THT vs. SMD
 - Power ratings (1/8 W, 1/4 W, 1/2 W, 1 W, 5 W, 10 W)
 - Tolerance (1%, 5%, 10%, etc.)
 - Type (fixed, variable, carbon film, metal film, wire-wound, etc.)Include part numbers, prices, and local suppliers.
- Identify five special-purpose resistors (e.g., thermistors, LDRs, varistors). Explain their purpose, circuit applications, and local availability with part numbers and suppliers.
- Compare the multimeters Brother HD-470A, Sunshine DT-890N, and UNI-T UT33A+ in terms of specifications, features, accuracy, safety, price, and local availability. Recommend the best model with justification.

Capacitor (C)



Online Tutorial on Capacitor:

https://www.electronics-tutorials.ws/capacitor/cap_1.html

► **1. Introduction to Capacitors**

2. Types of Capacitor

3. Capacitor Characteristics

4. Capacitance and Charge

5. Capacitor Colour Codes

6. Capacitors in Parallel

7. Capacitors in Series

8. Capacitance in AC Circuits

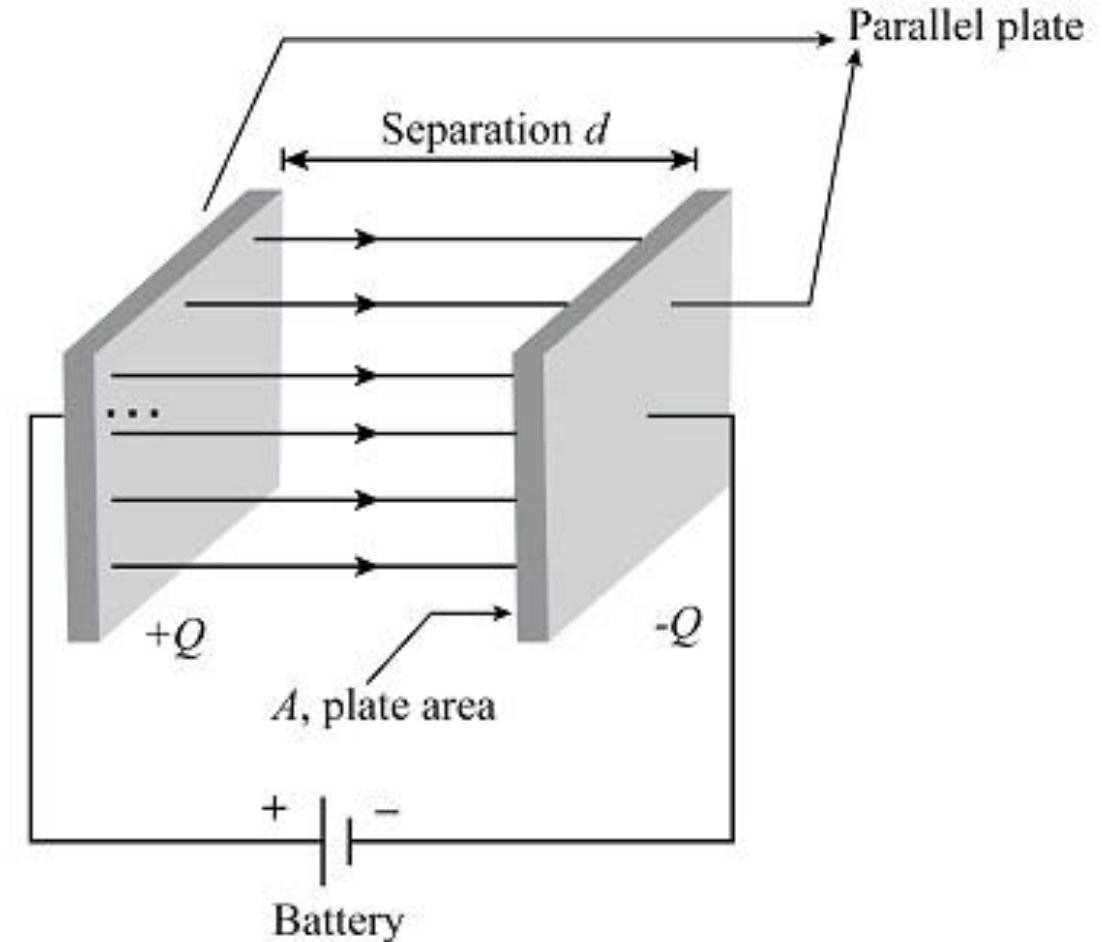
9. Capacitor Tutorial Summary

10. Capacitive Voltage Divider

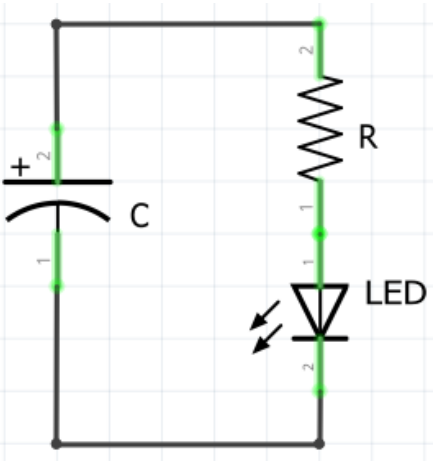
11. Ultracapacitors

What is a Capacitor?

- A **capacitor** is a passive electronic component that stores electrical energy temporarily in the form of an **electric field**.
- It has two conductive plates separated by an insulator (dielectric).
- A voltage source forces electrons onto one plate and removes electrons from the other.
- The resulting electric field between plates stores the energy.

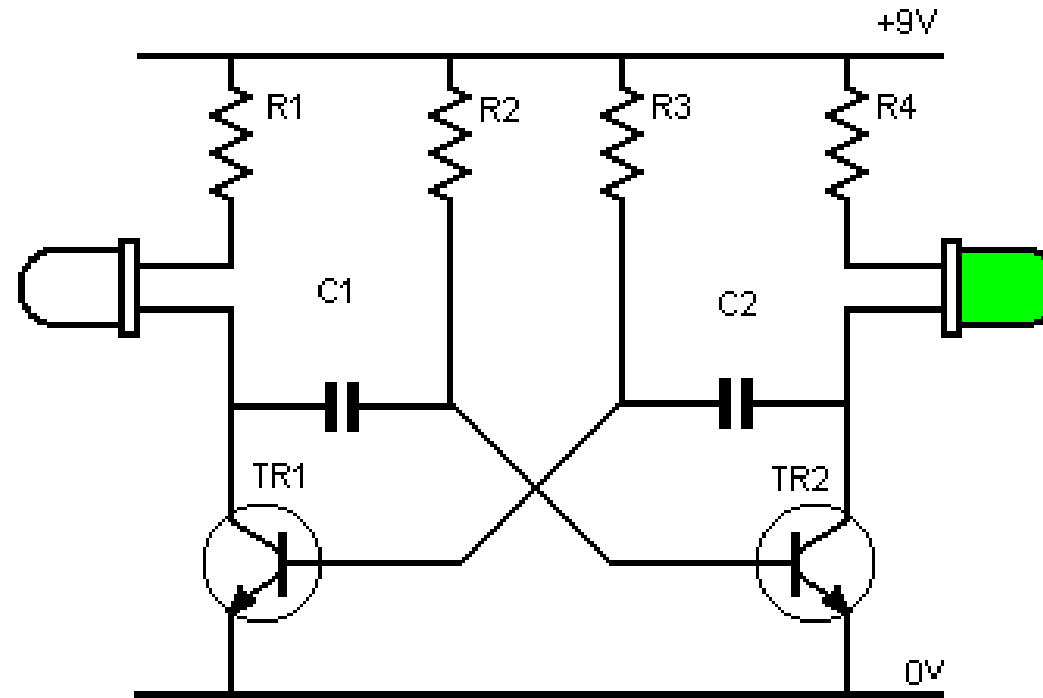


Check the Capacity



Purpose of Capacitor

- Store charge
- Filter noise
- Smooth power supply
- Timing circuits
- AC coupling
- Etc.



Units of Capacitance

- The SI unit of electrical capacitance is **Farad (F)**.
- Prefixes:

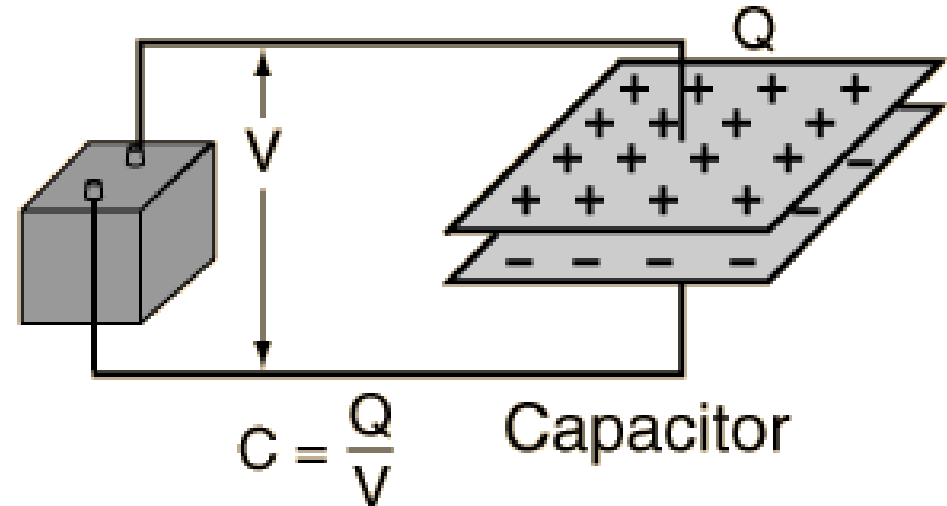
Unit	Multiplier	Example
Farad (F)	1	1 F
Milifarad (mF)	10^{-3}	12 mF = 0.012 F
Microfarad (μ F)	10^{-6}	12 μ F = 0.000012 F
Nanofarad (nF)	10^{-9}	12 nF = 0.000000012 F
Picofarad (pF)	10^{-12}	12 pF = 0.000000000012 F

Capacitance Formula

- Capacitance is defined as:

$$C = \frac{Q}{V}$$

- C = Capacitance (Farads)
 - Q = Charge (Coulombs)
 - V = Voltage (Volts)
- E.g., If a 5V is applied to a 100 μF (100×10^{-6} F) capacitor, then the charge it stored = $100 \times 10^{-6} \times 5 = 0.0005 \text{ C}$ (0.5 mC)

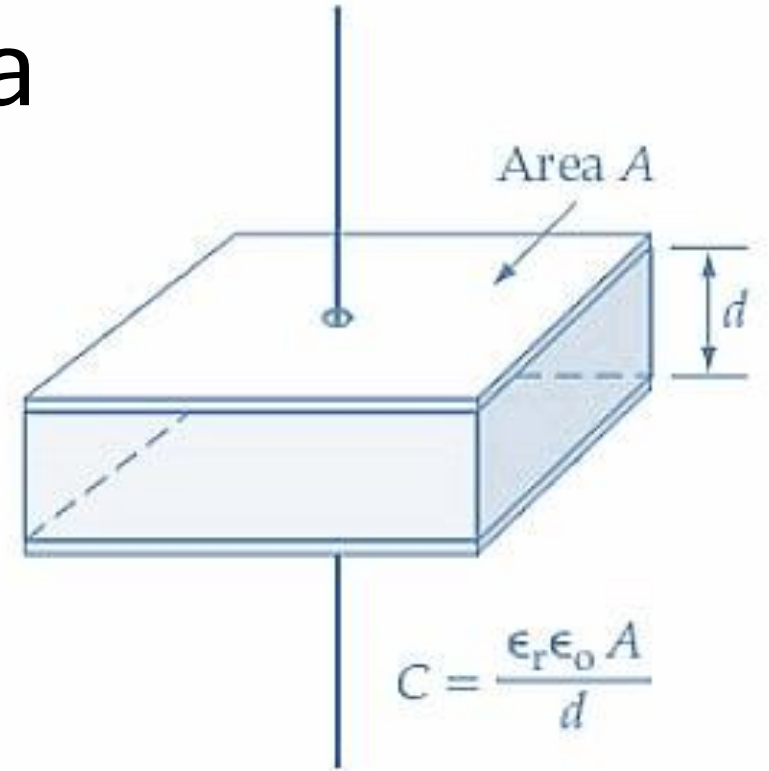


Parallel Plate Capacitor Formula

- The capacitance of a parallel plate capacitor is determined by:

$$C = \frac{\epsilon A}{d} \qquad C = \frac{\epsilon_0 \epsilon_r A}{d}$$

- C** = Capacitance (Farads, F)
- ϵ** = Permittivity of dielectric material
- A** = Area of one plate (m²)
- d** = Distance between plates (m)
- ϵ_0** = Permittivity of free space $\approx 8.854 \times 10^{-12}$ F/m
- ϵ_r** = Relative permittivity (dielectric constant)



How Much Energy is Stored?

- The energy E (in joules) stored in a capacitor is:

$$E = \frac{1}{2}CV^2$$

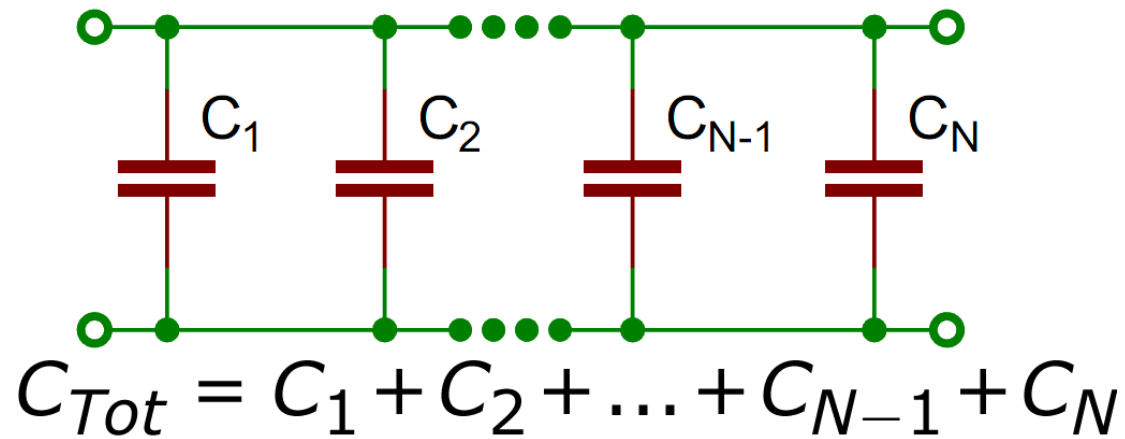
Capacitor Configurations

- Capacitors in Series:



$$\frac{1}{C_{Tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_{N-1}} + \frac{1}{C_N}$$

- Capacitors in Parallel:

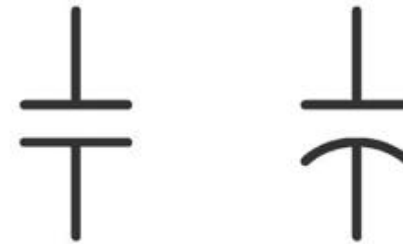


Polarized vs Non-Polarized

- Polarized Capacitors
 - Must connect correctly.
 - E.g., Electrolytic and Tantalum
- Non-Polarized Capacitors
 - Can connect either direction.
 - E.g., Ceramic, Film and Mica



Non-polarized
Capacitor



Polarized
Capacitor



Capacitor Characteristics

- **Capacitance**

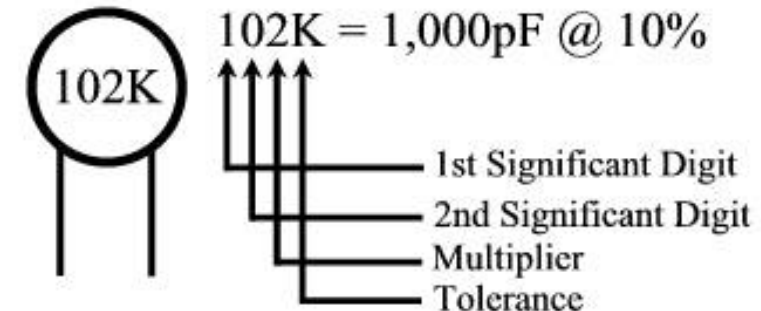
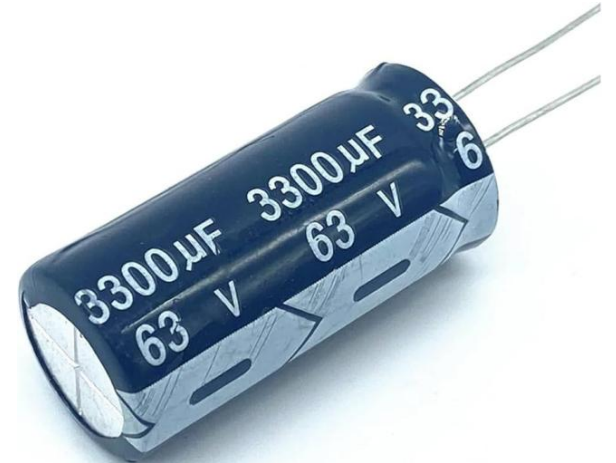
- The amount of electric charge a capacitor can store, e.g., 10 μ F.

- **Voltage Rating**

- The **maximum safe voltage** that can be applied across the capacitor.
- E.g., 10V, 16V, 25V, 35V, 50V, 63V, 100V, 160V, 250V, 400V and 1000V
- Use 1.5 $\times$ to 2 $\times$ higher voltage rating than actual circuit voltage.

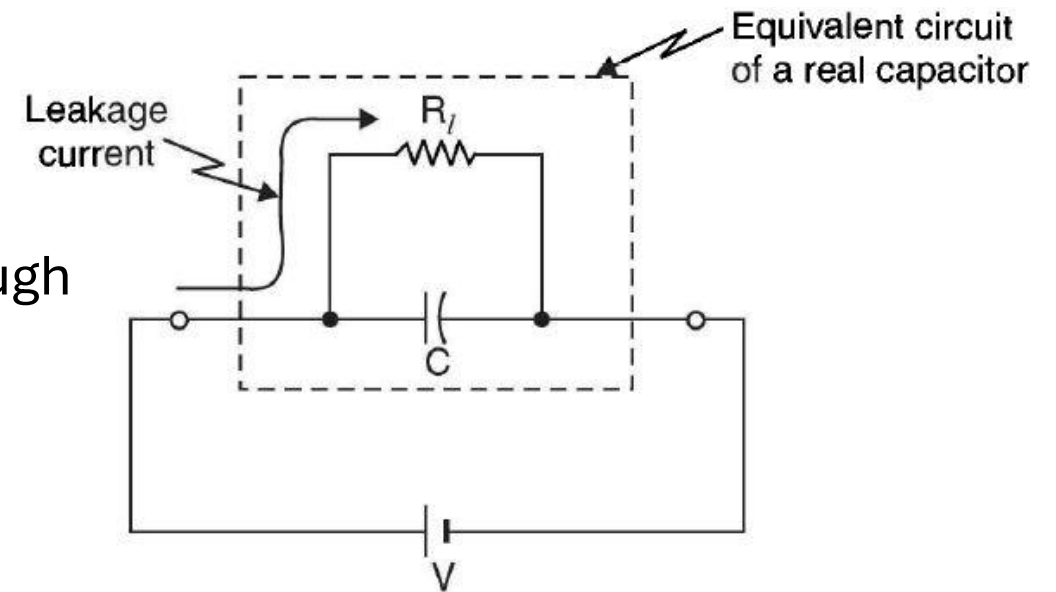
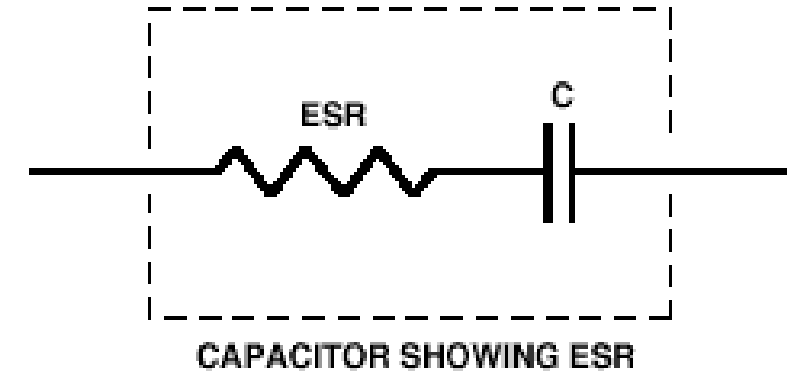
- **Tolerance**

- Indicates how much the actual capacitance may vary from the labeled value.
- E.g., 1%, 5% and 10%



Capacitor Characteristics Cont.

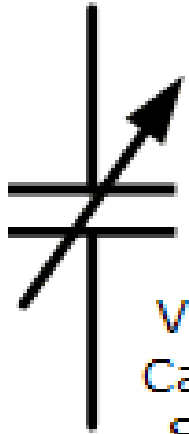
- **Polarization**
- **ESR (Equivalent Series Resistance)**
 - A real capacitor behaves like an ideal capacitor + a small internal resistance or ESR.
- **Leakage Current**
 - Ideally capacitors block DC current. Real capacitors allow a tiny current to leak through dielectric.
- **Temperature Coefficient**
 - Capacitance changes with temperature.
- Etc.



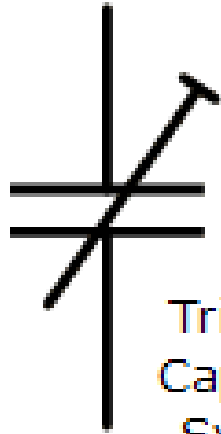
Types of Capacitors

- Variable Capacitors
- Electrolytic Capacitors
- Ceramic Capacitors
- Film (Mylar) Capacitors
- Tantalum Capacitors
- Supercapacitors
- Mica Capacitors
- THT and SMD Capacitors
- Etc.

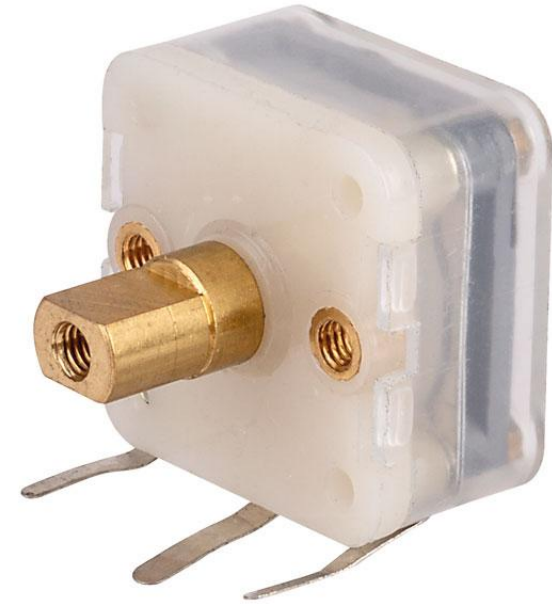
Variable Capacitors



Variable
Capacitor
Symbol

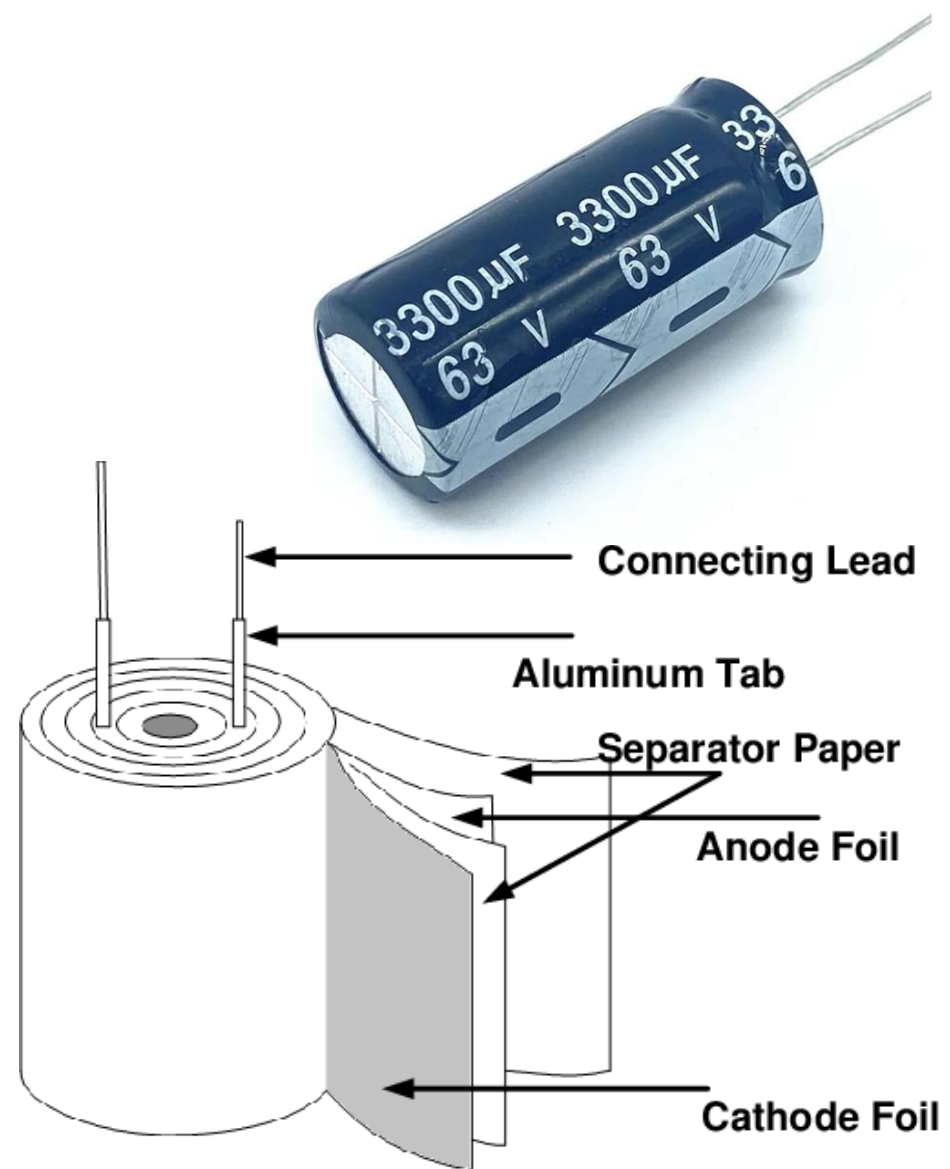


Trimmer
Capacitor
Symbol



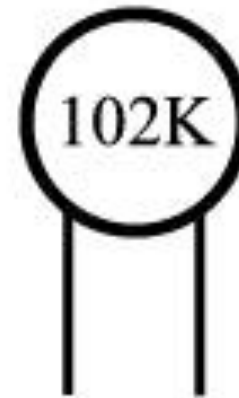
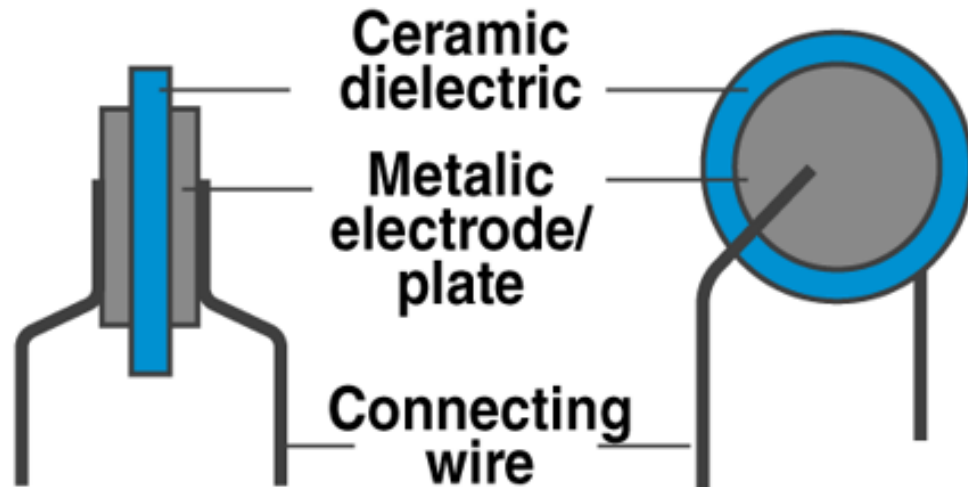
Electrolytic Capacitors

- A high-capacitance polarized capacitor that uses an electrolytic solution and an oxide layer as the dielectric.
- Uses: power supply filtering, audio circuits



Ceramic Capacitors

- A small non-polarized capacitor that uses a ceramic material as the dielectric.
- Uses: decoupling, RF circuits



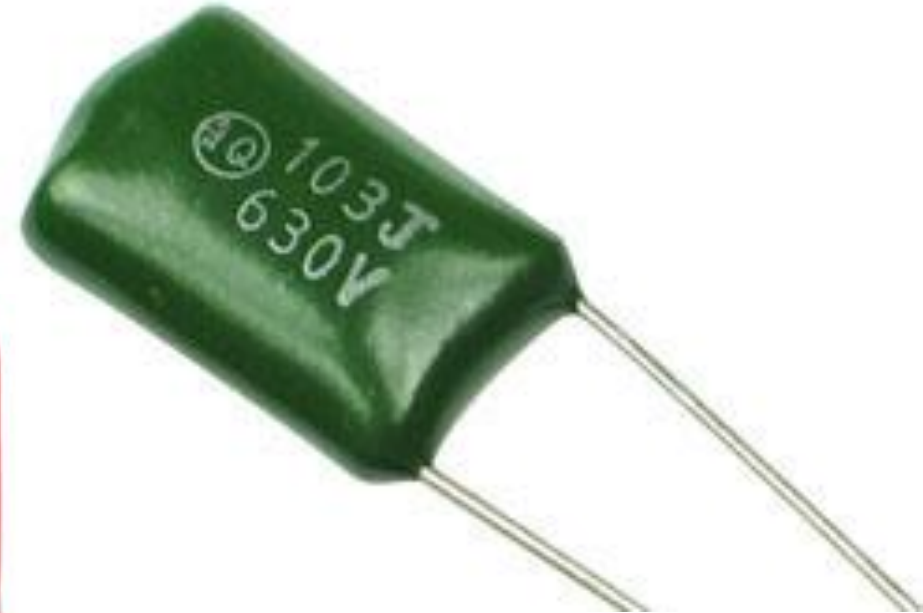
$102K = 1,000\text{pF} @ 10\%$

↑ ↑ ↑
1st Significant Digit
2nd Significant Digit
Multiplier
Tolerance



Film (Mylar) Capacitors

- Uses a plastic film dielectric such as polyester, polypropylene, or PTFE.
- Uses: audio, timing, high voltage



Other Capacitors

- **Tantalum Capacitors**

- Small size, stable, polarized, expensive
- Used in precision circuits

- **Supercapacitors**

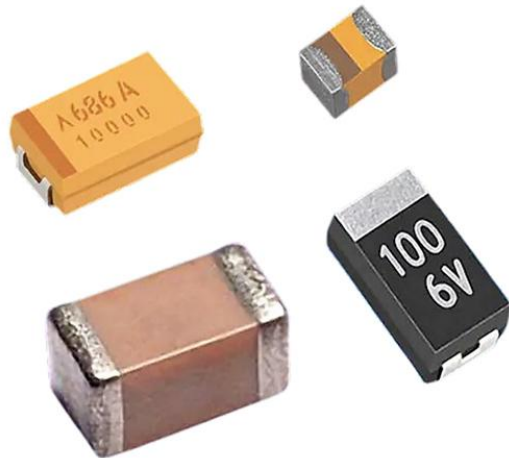
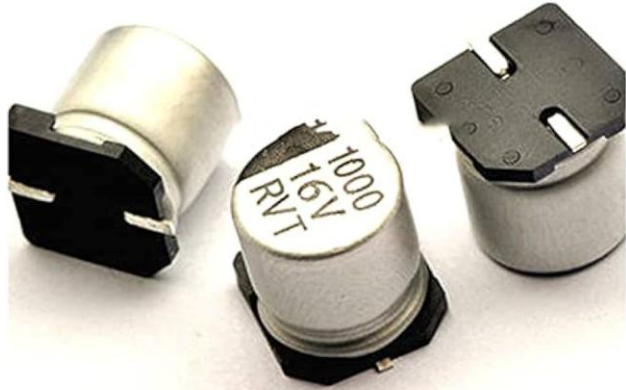
- Extremely high capacitance, fast charging, backup power, energy storage

- **Mica Capacitors**

- Very stable, RF use, expensive



SMD Capacitors



Capacitor Time Constant, Charging and Discharging Equations

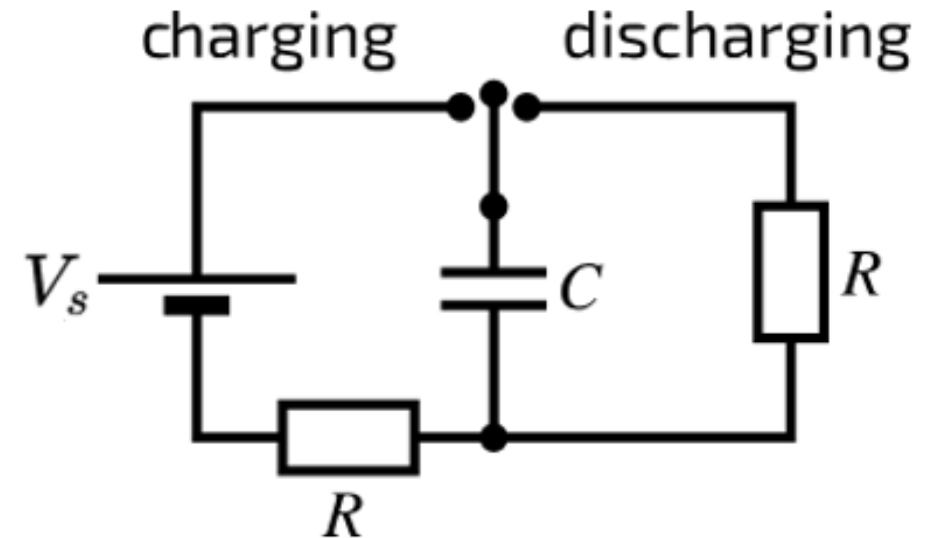
- Charging Equation

$$V_C(t) = V_s \left(1 - e^{-t/(RC)} \right)$$

- Discharging Equation

$$V_C(t) = V_0 e^{-t/(RC)}$$

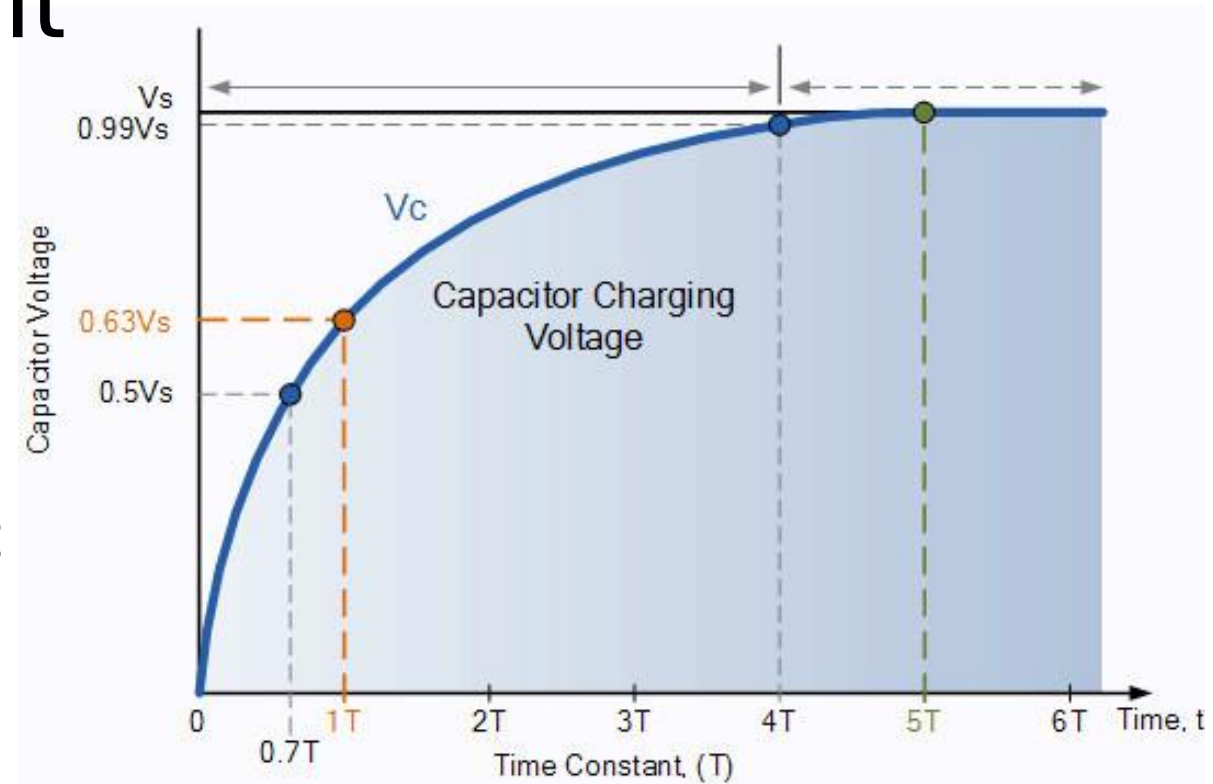
- Time Constant (τ , tau) $\tau = RC$
 - The RC time constant determines charging/discharging speed.



- $V_c(t)$ = Voltage across the capacitor at time t
- V_s = Source voltage
- V_0 = Voltage across the capacitor once charged
- R = resistance
- C = capacitance

Capacitor Time Constant

- E.g., $R=1\text{k}\Omega$ and $C=100\mu\text{F} \rightarrow \tau = 1000 \times 100 \times 10^{-6} = 0.1$ seconds
- The time constant represents the time it takes for:
 - A charging capacitor to reach about 63% of the final voltage.
 - A discharging capacitor to drop to about 37% of its initial voltage.
- After 5τ , the capacitor is considered fully charged or discharged (~99%).



Exercise

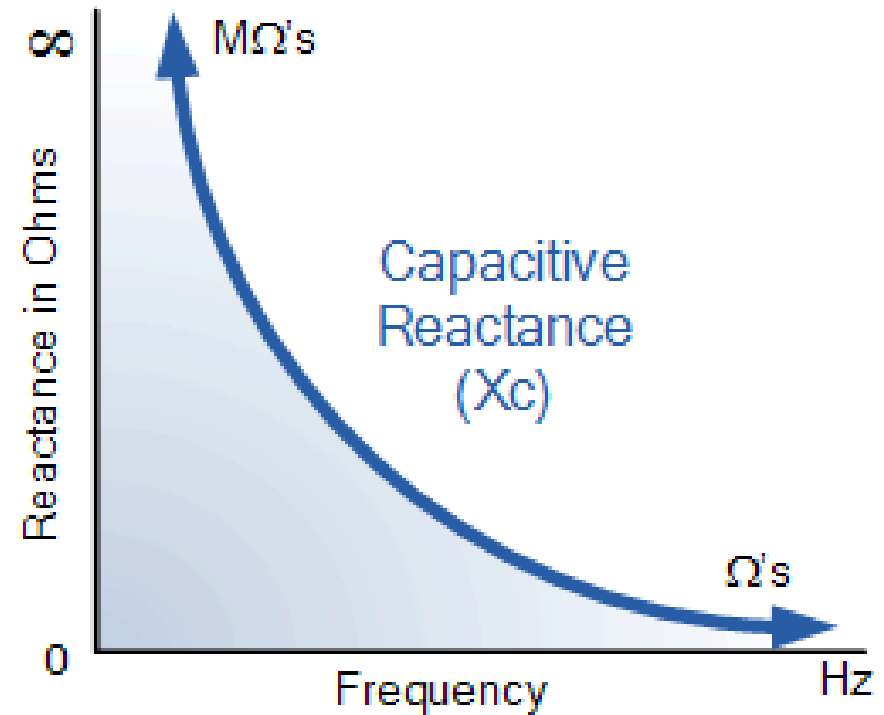
- Q1) A $100\mu F$ capacitor is connected across a $12V$ battery. Calculate the charge stored on the capacitor.
- Q2) A $50\mu F$ capacitor is charged to $20V$. Calculate the energy stored.
- Q3) A $220\mu F$ capacitor is connected in series with a $1k\Omega$ resistor. Calculate the time constant of the circuit.
- Q4) A $10\mu F$ capacitor is charging through a $1k\Omega$ resistor from a $5V$ battery. Find the voltage across the capacitor after $t=0.01s$.
- Q5) Calculate the total capacitance of two capacitors - $6\mu F$ and $3\mu F$ - when they are connected in series and in parallel.
- Q6) A $100\mu F$ capacitor is charging through a $1k\Omega$ resistor from a $10V$ battery. Find the voltage across the capacitor after one time constant.
- Q7) A $220\mu F$ capacitor is charging through a 470Ω resistor from a $12V$ supply. Calculate the time it takes for the capacitor to reach 90% of the supply voltage.
- ANS: Q1) $1.2mC$, Q2) $0.01J$, Q3) $0.22s$, Q4) $\approx 3.16V$, Q5) $2\mu F$ and $9\mu F$, Q6) $\approx 6.32V$, Q7) $\approx 0.238s$

Capacitive Reactance

- Capacitive reactance (similar to resistance) is the opposition offered by a capacitor to alternating current (AC), given by the formula:

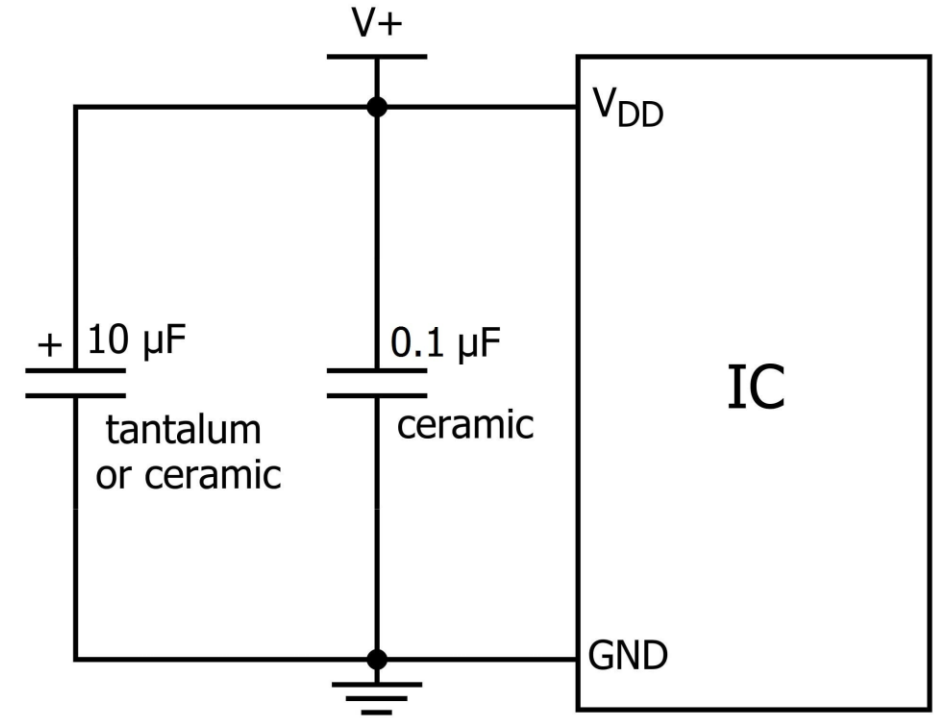
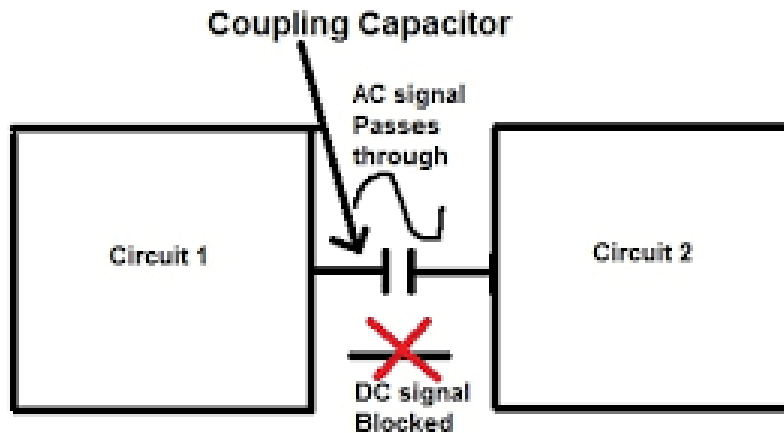
$$X_C = \frac{1}{2\pi fC}$$

- X_C = Capacitive Reactance (Ω)
 - f = Frequency (Hz)
 - C = Capacitance (F)
- A capacitor allows high-frequency signals to pass more easily. It blocks DC.



Capacitor Applications

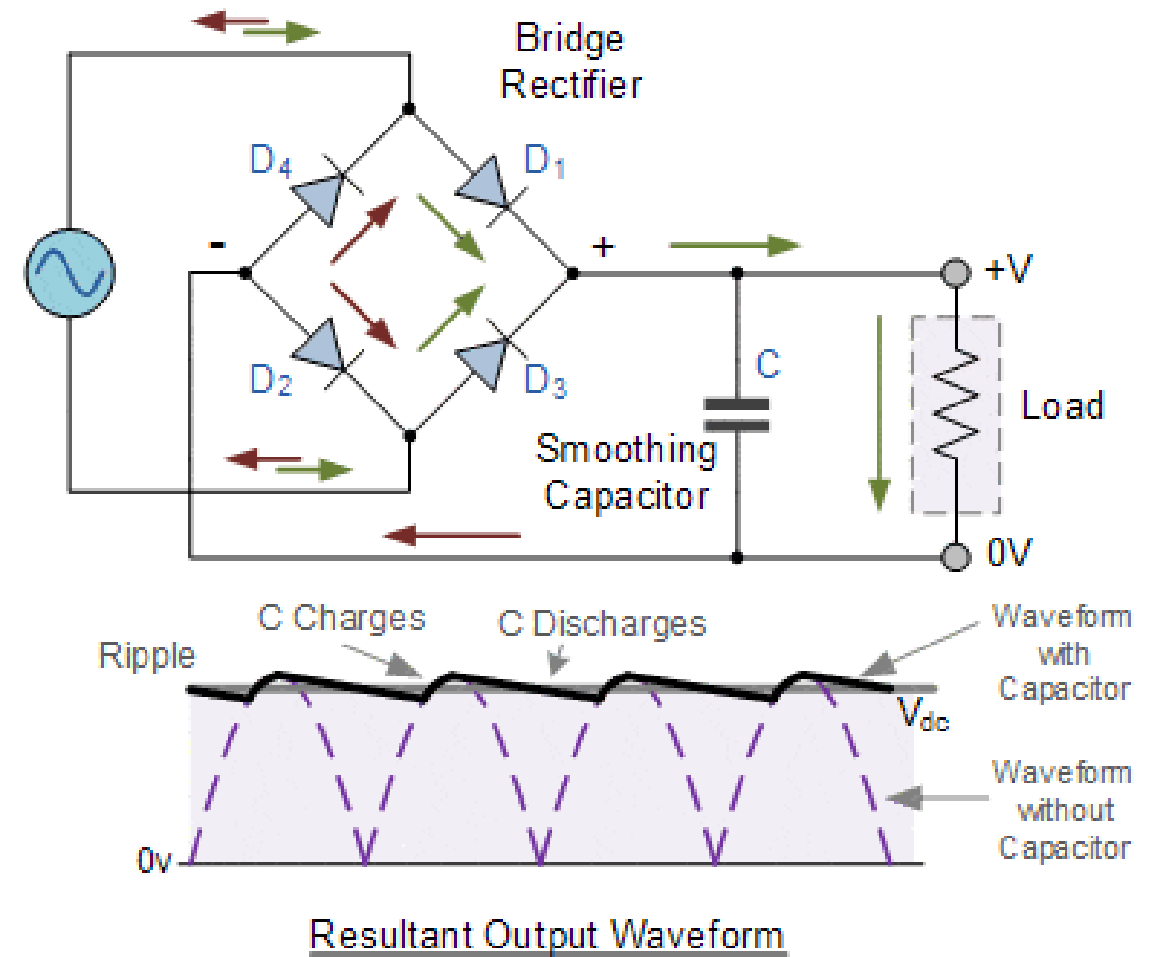
- **AC Coupling (DC Blocking)**
 - E.g., audio amplifiers, microphone circuits, communication systems
- **Decoupling / Bypass Capacitors**
 - Used near IC power pins to remove high-frequency noise and stabilize supply voltage



Capacitor Applications Cont.

- **Power Supply Filtering**

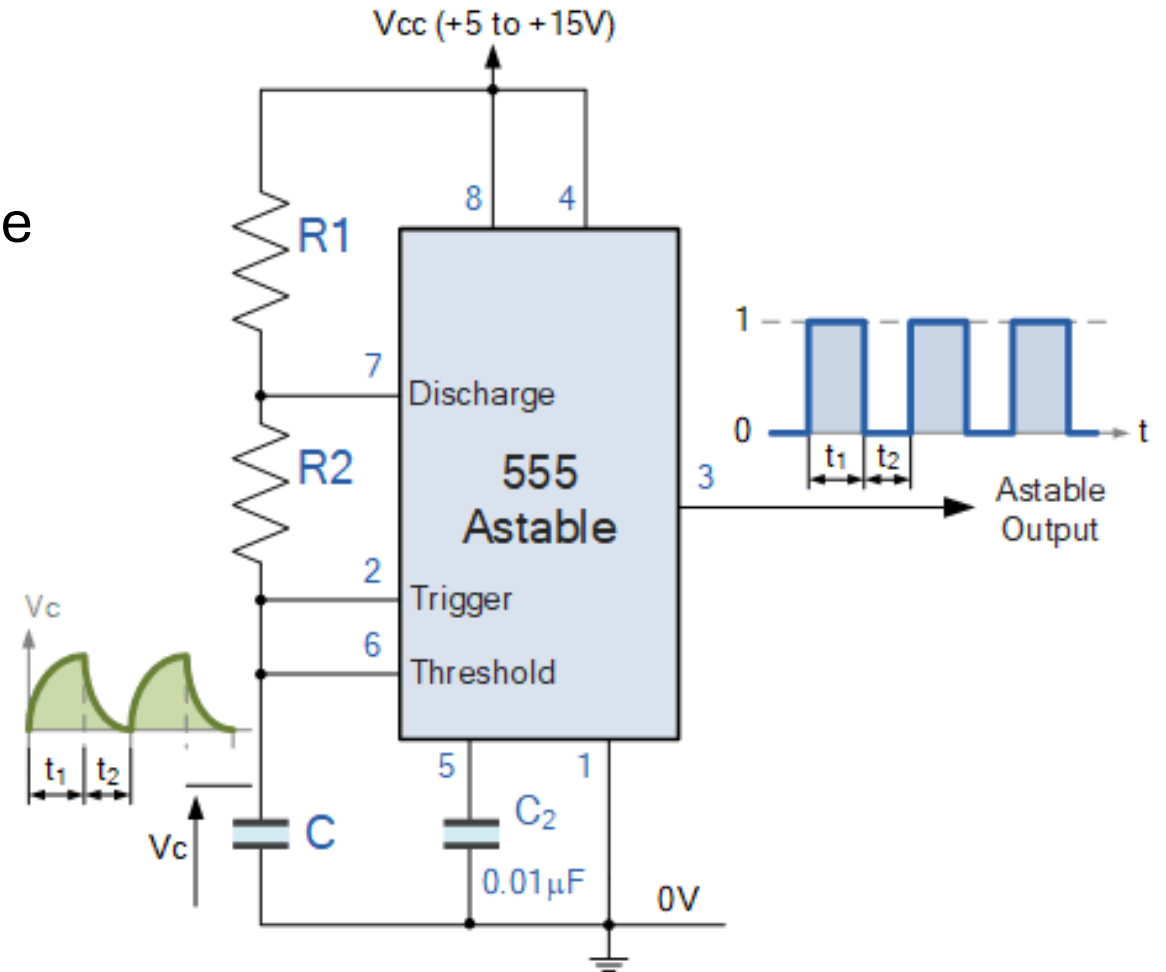
- Used after rectifiers to smooth pulsating DC and to reduce ripple voltage



Capacitor Applications Cont.

- **RC Timing Circuits**

- E.g., delays, oscillators, timers, pulse generation



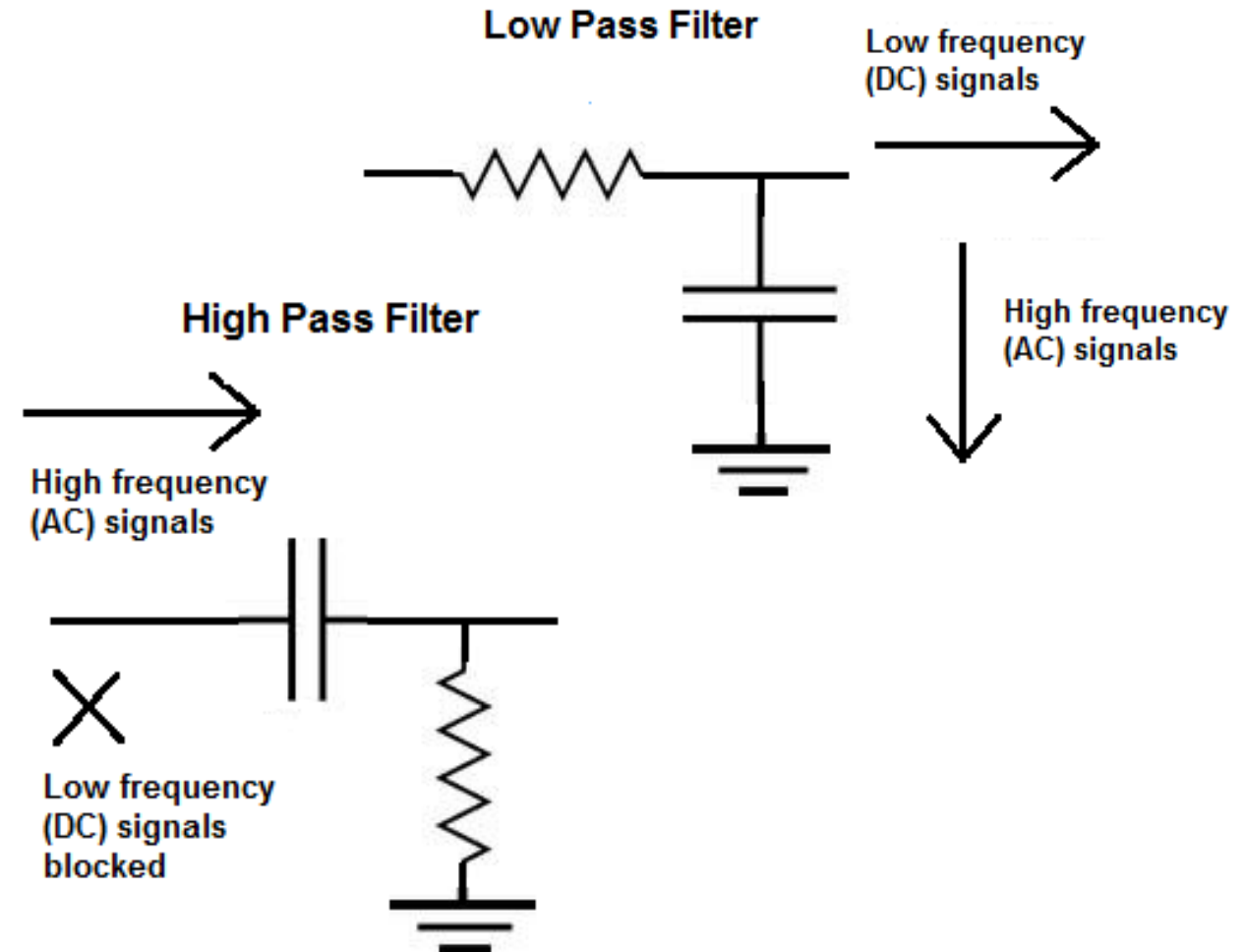
Capacitor Applications Cont.

- **Low-Pass Filters**

- Allows low frequencies to pass.
- E.g., audio tone control, noise reduction, analog filtering

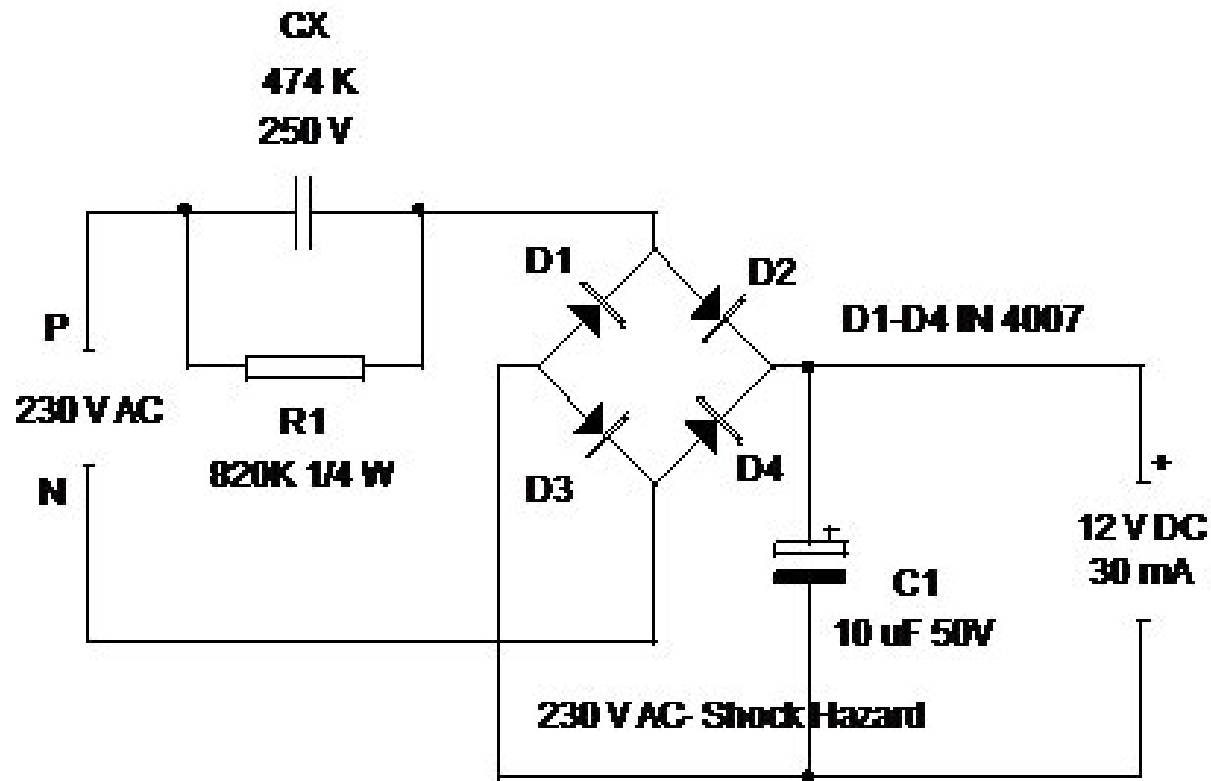
- **High-Pass Filters**

- Allows high frequencies to pass.
- E.g., audio crossover, signal conditioning, RF systems



Capacitor Applications Cont.

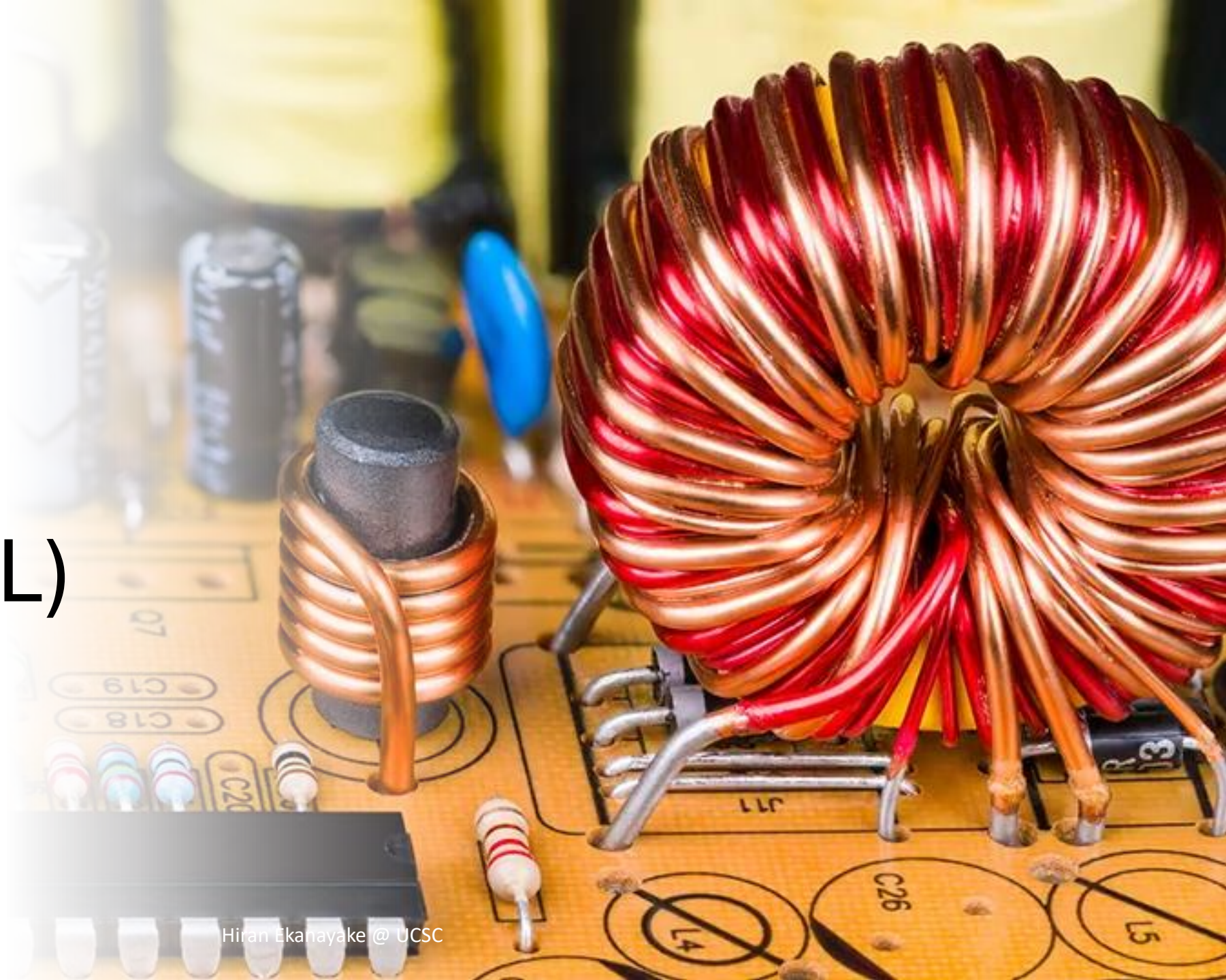
- Transformerless Power Supply



Activity Cont.

- Investigate the local availability and prices of 0.1uF and 100uF capacitors. Compare prices based on:
 - THT vs. SMD
 - Voltage ratings (10V, 16V, 25V, 35V, 50V, 63V, 100V, 160V, 250V, 400V, etc.)
 - Tolerance (1%, 5%, 10%, etc.)
 - Type (electrolytic, film, ceramic, tantalum, etc.)
- Include part numbers, prices, and local suppliers.

Inductor (L)



Online Tutorial on Inductor:

<https://www.electronics-tutorials.ws/inductor/inductor.html>

- ▶ 1. The Inductor

- 2. Inductance of a Coil

- 3. Mutual Inductance

- 4. Inductors in Series

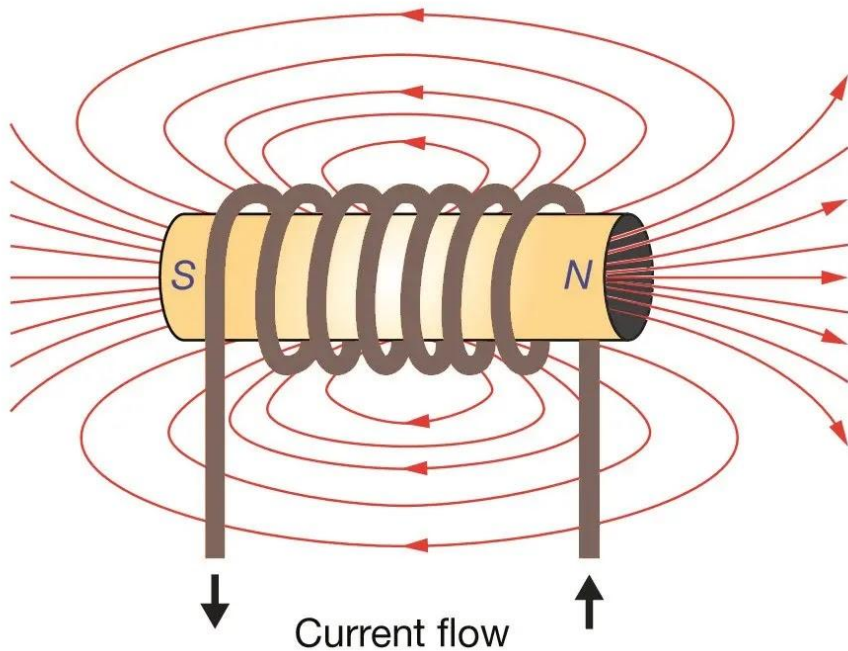
- 5. Inductors in Parallel

- 6. LR Series Circuit

- 7. Inductive Reactance

What is an Inductor?

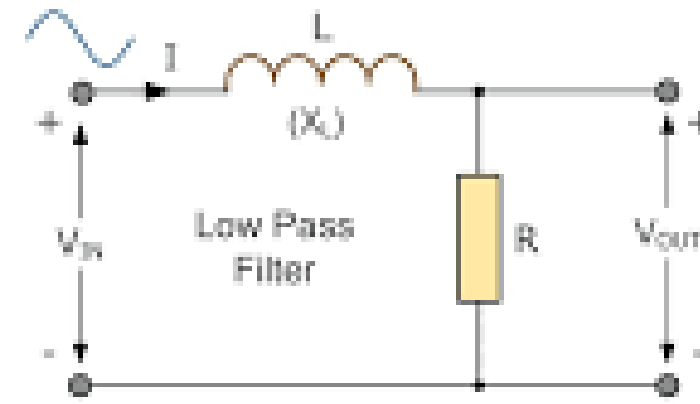
- An **inductor** (or choke) is a passive electronic component that stores energy in a **magnetic field** when electric current flows through it.



RL FILTERS

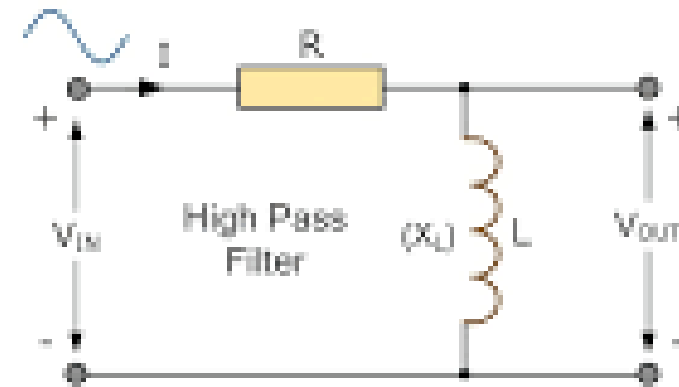
Purpose of Inductor

- Filters
- SMPS (Switched-Mode Power Supply)
- RF circuits
- Energy storage
- Etc.



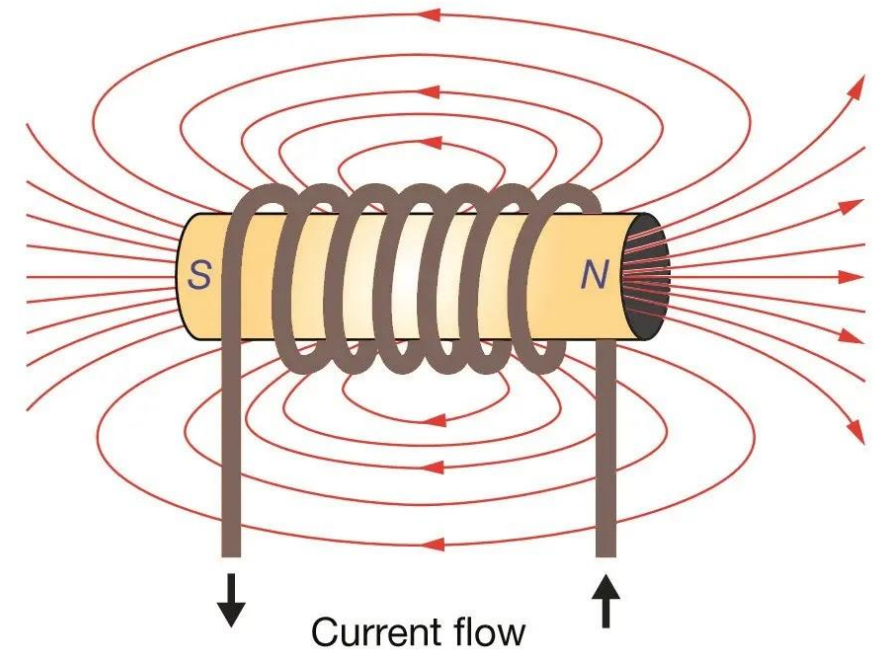
$$f_c = \frac{R}{2\pi L} \text{ (Hz)} \quad X_L = 2\pi fL$$

© ElectronicsRD



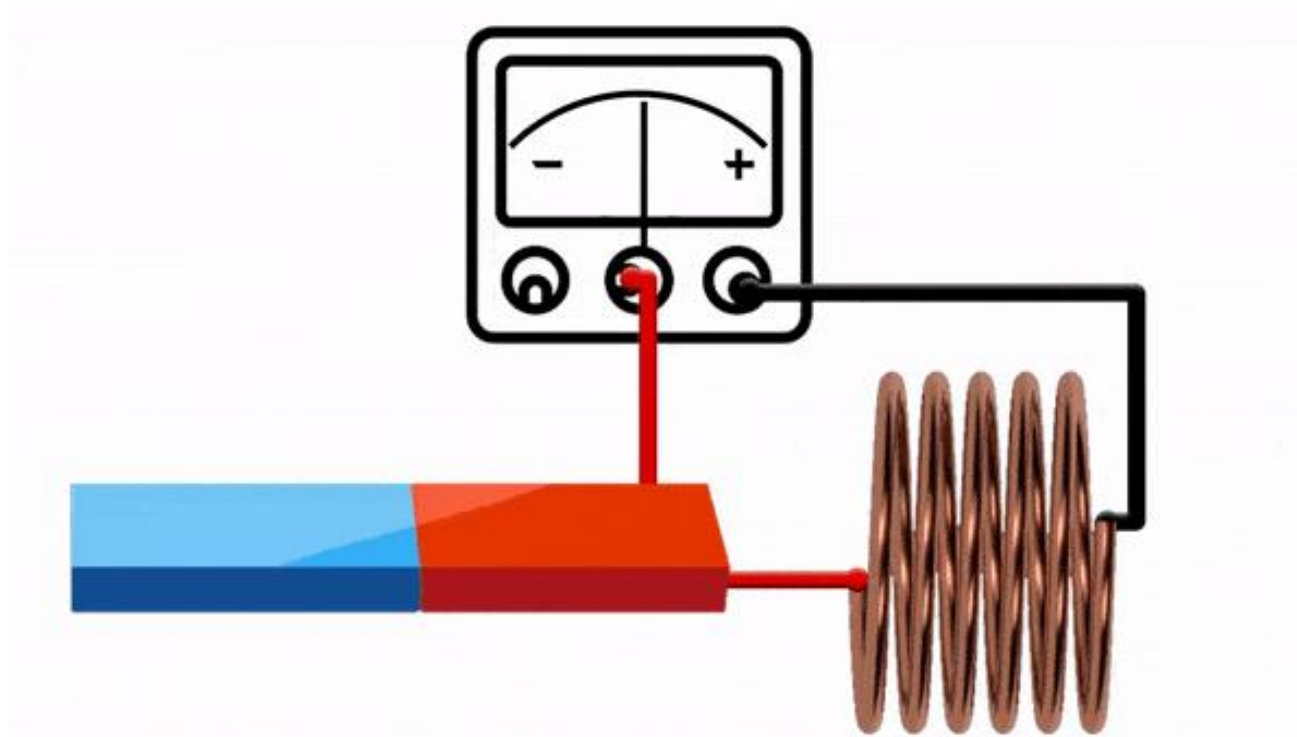
How an Inductor Works?

- When current flows through a coil:
 - A magnetic field is created
 - Increasing current strengthens the field
 - Decreasing current collapses the field
 - The collapsing field induces voltage opposing the change
- This is based on Faraday's Law, Lenz's Law, etc.



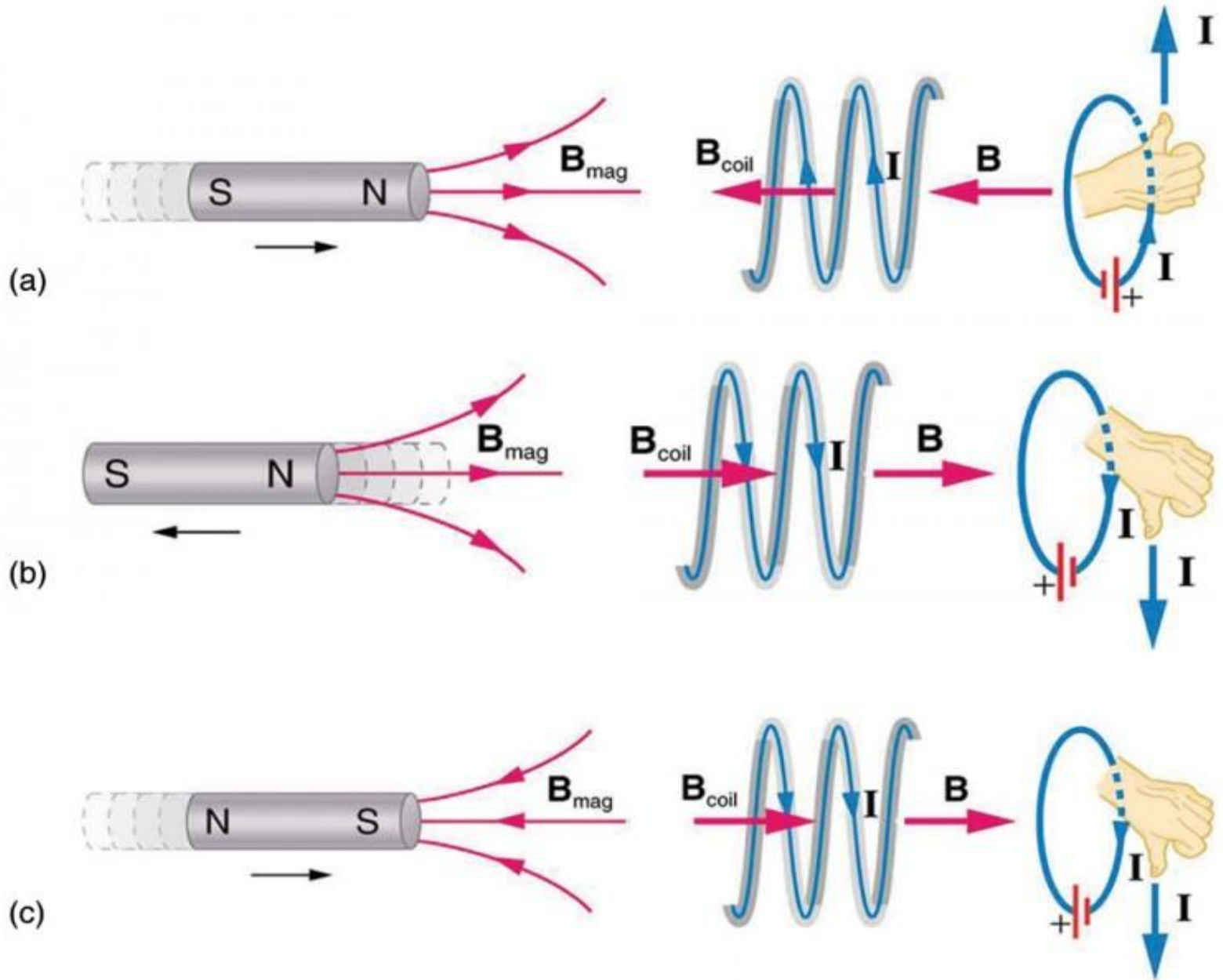
Faraday's Law of Electromagnetic Induction

- A changing magnetic field in a closed loop/circuit induces an **electromotive force (EMF)** in the wire.



The Direction

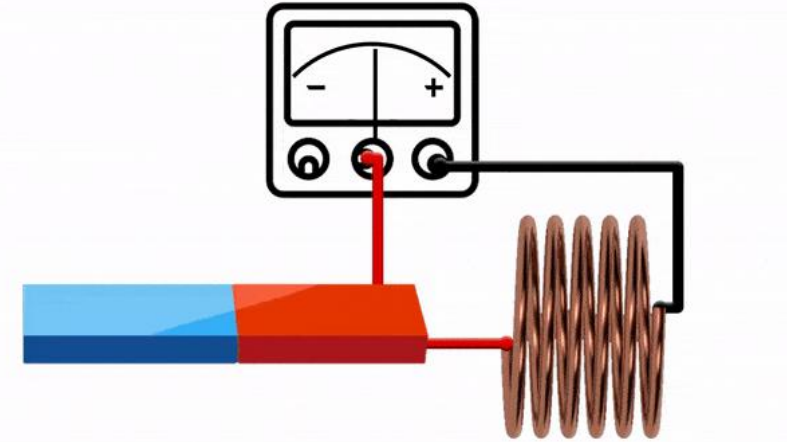
- **Lenz's Law** – induction opposes any change in flux



Faraday's Law of Electromagnetic Induction

- The coil has **N** turns.
- The negative sign represents **Lenz's Law**, indicating the direction of the induced emf.

$$e = -N \frac{d\Phi}{dt}$$



e = induced emf (volts)

Φ = magnetic flux (webers, Wb)

t = time (seconds)

Magnetic Flux

$$\Phi = BA \cos \theta$$

B = magnetic flux density (tesla, T)

A = area of the coil (m^2)

θ = angle between the magnetic field and the normal to the coil

Units of Inductance

- The SI unit of electrical inductance is **Henry (H)**.
- Prefixes:

Symbol	Abbreviation	Value in Numbers
mH	Milli-henry	10^{-3}
μH	Micro-henry	10^{-6}
nH	Nino-henry	10^{-9}



Self-Inductance

- When the current in a coil changes, the coil opposes that change by inducing its own emf.
- A circuit has an inductance of **1 henry (H)** if a current change of 1 ampere per second induces an emf of 1 volt.

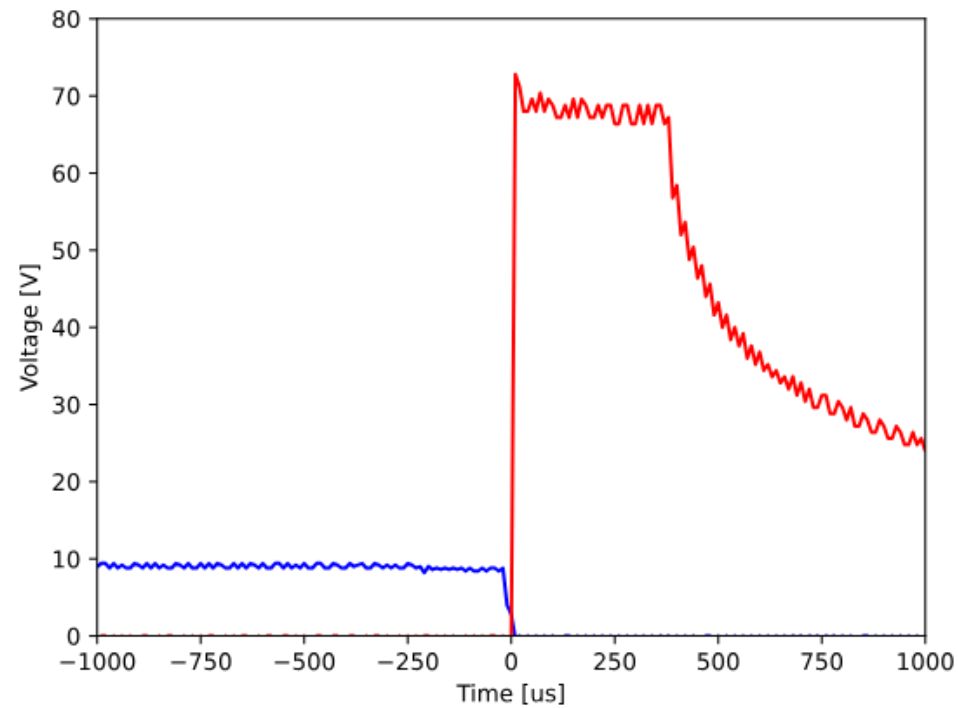
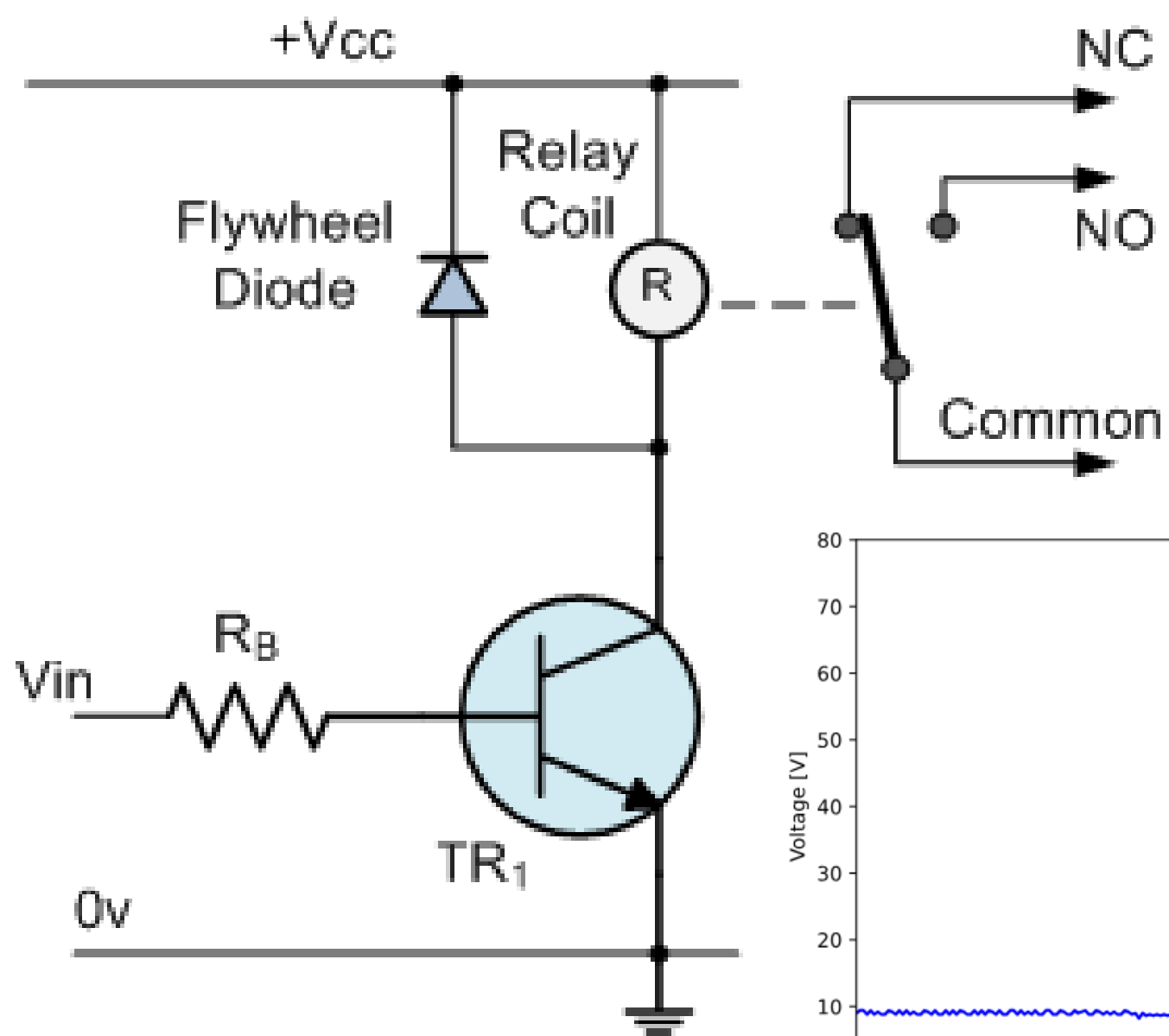
$$L = \frac{N\Phi}{I}$$

N = number of turns,

Φ = magnetic flux (Wb),

I = current (A).

$$e = -L \frac{dI}{dt}$$



Exercise

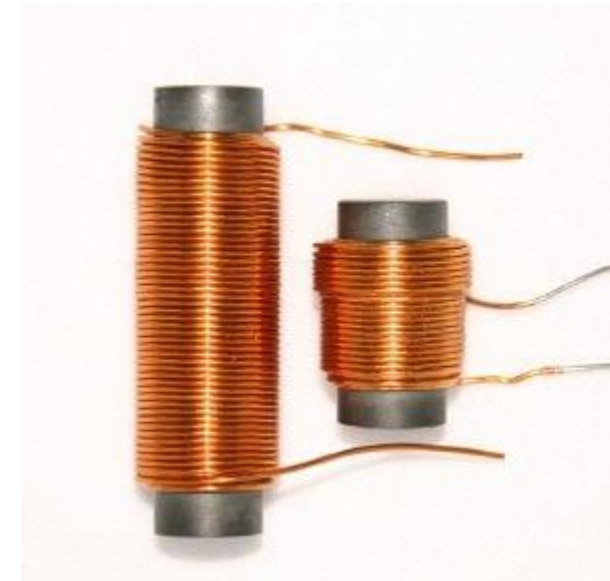
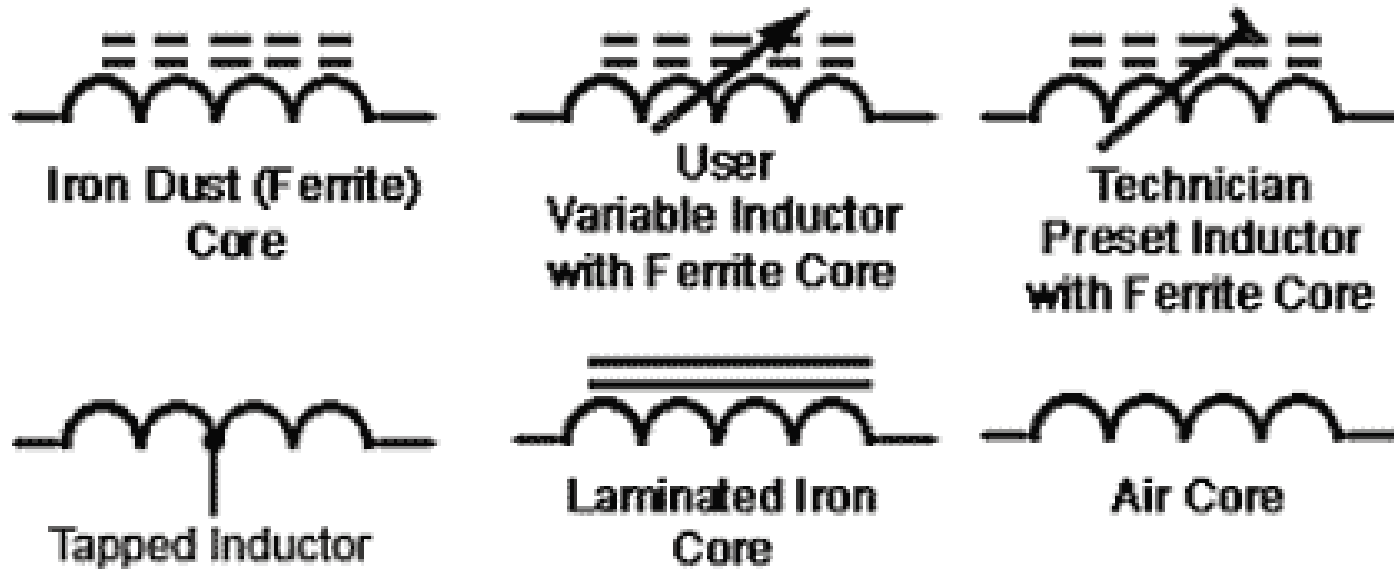
- Q1) A coil has a self-inductance of 4 H . The current through it increases uniformly from 1 A to 5 A in 0.5 s . Calculate the induced emf.
- Q2) An emf of 10 V is induced in a coil when the current changes at a rate of 5 A/s . Find the self-inductance of the coil.
- Q3) A coil of 250 turns carries a current of 4 A and produces a magnetic flux of 1.6 mWb . Calculate the self-inductance of the coil.
- Q4) The current in a 3 H inductor decreases uniformly from 6 A to zero in 0.2 s . Calculate the induced emf.
- ANS: Q1) 32 V , Q2) 2 H , Q3) 0.1 H , Q4) 90 V

Types of Inductors

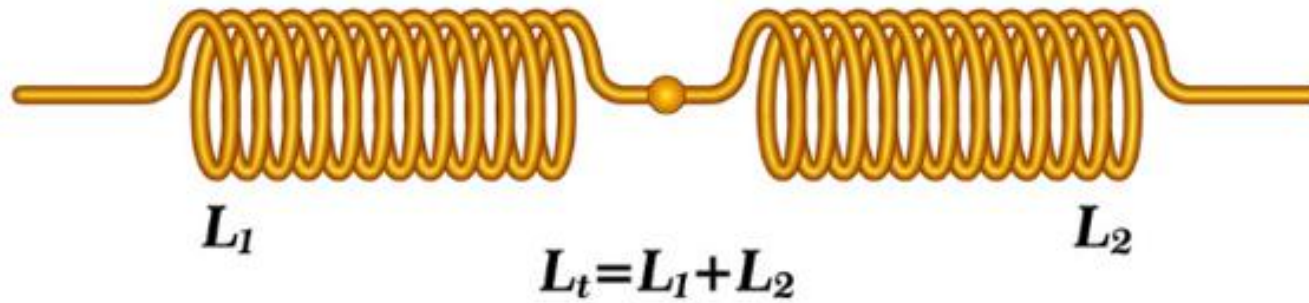
- Fixed Inductor
- Variable Inductor
- Power Inductor
- RF Inductor
- Choke
- Coupled Inductor
- Toroidal Inductor
- SMD Inductor



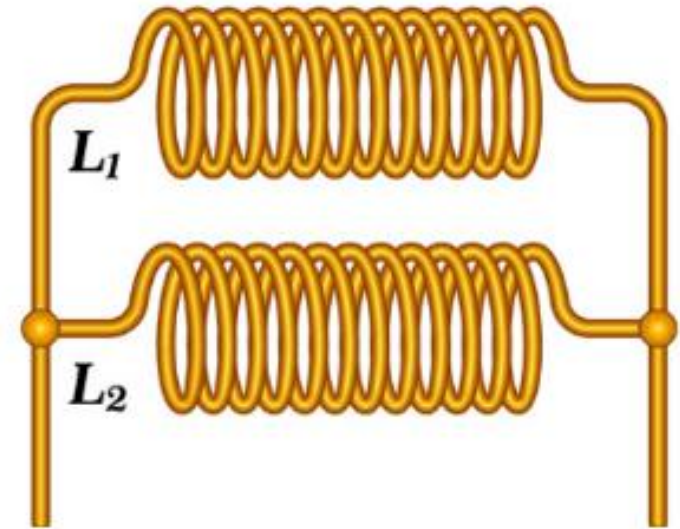
Inductor Symbols



Inductor Configurations



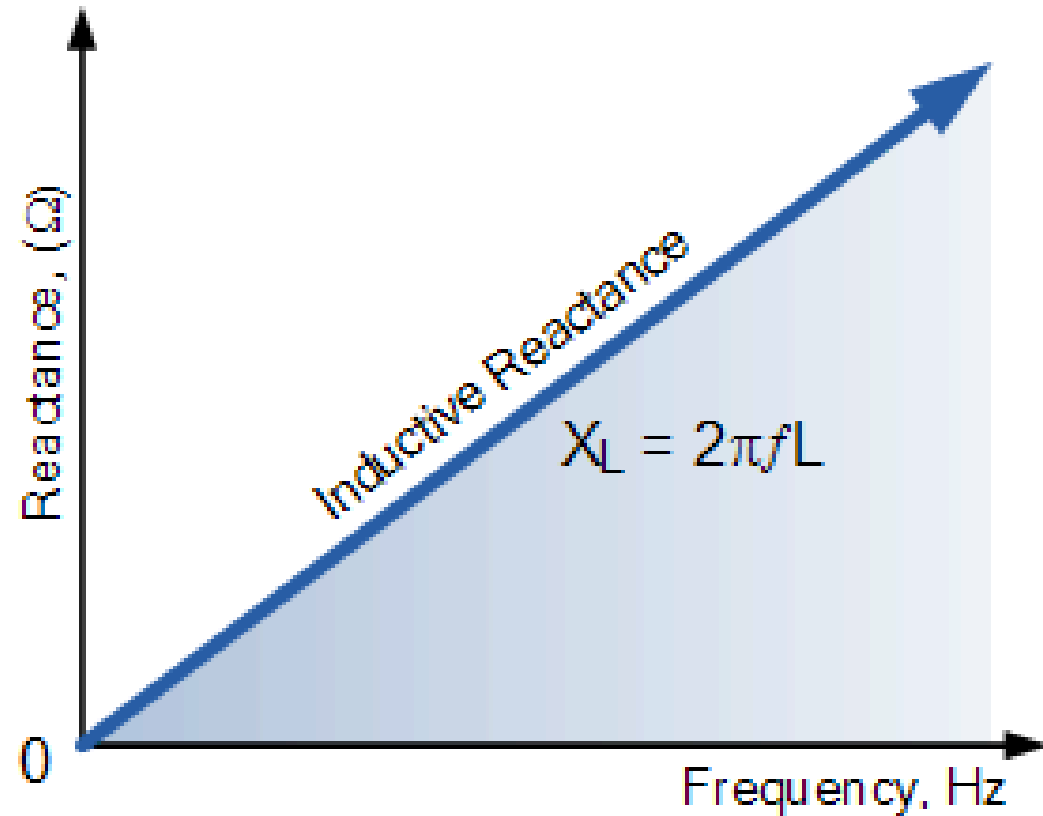
$$\frac{1}{L_t} = \frac{1}{L_1} + \frac{1}{L_2}$$



Inductive Reactance

- The opposition offered by an inductor to the flow of **alternating current (AC)** due to the **self-induced emf** produced in the inductor when the current changes.

$$X_L = 2\pi f L$$

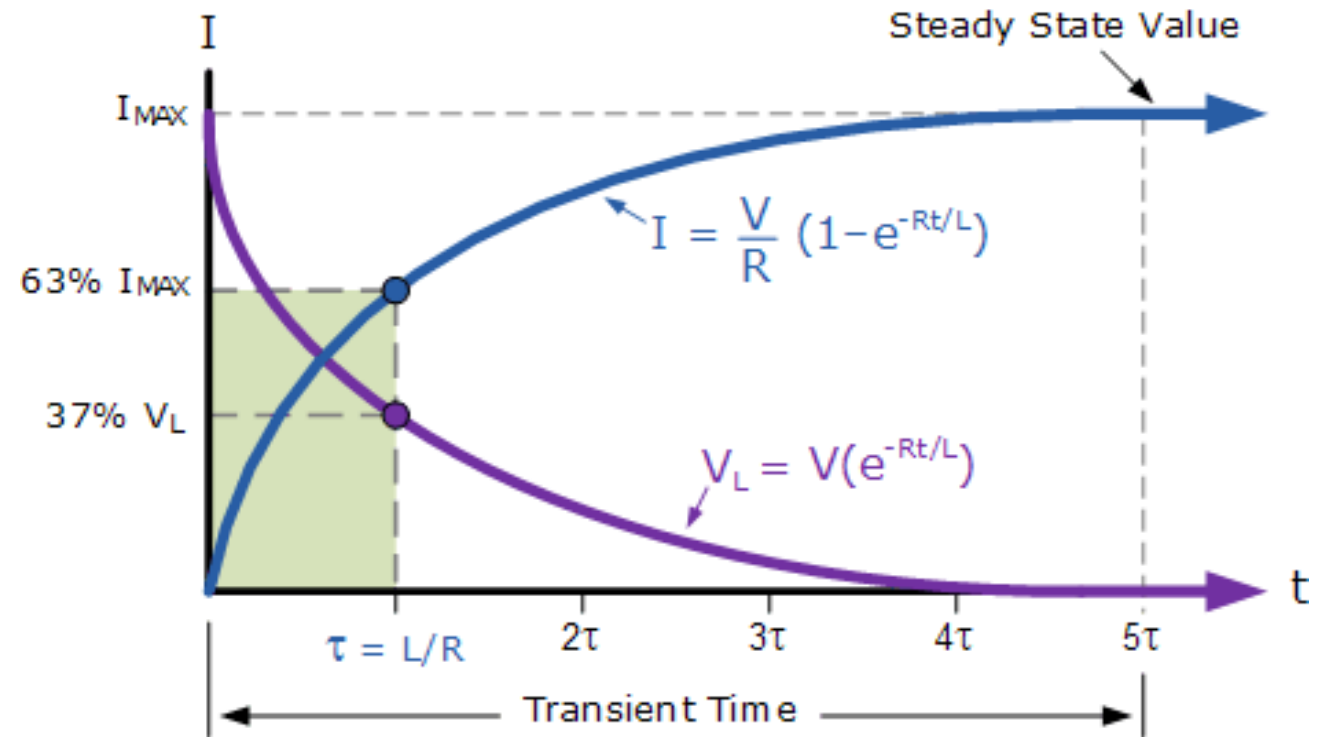
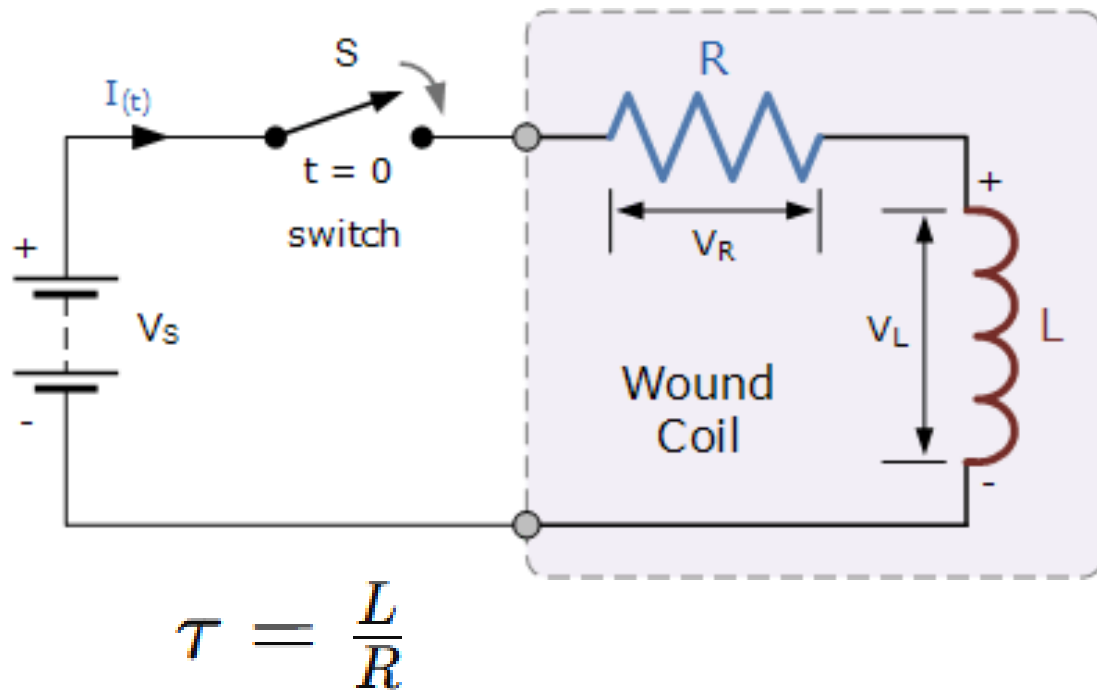


X_L = inductive reactance (ohms, Ω)

f = frequency of AC supply (Hz)

L = inductance (henry, H)

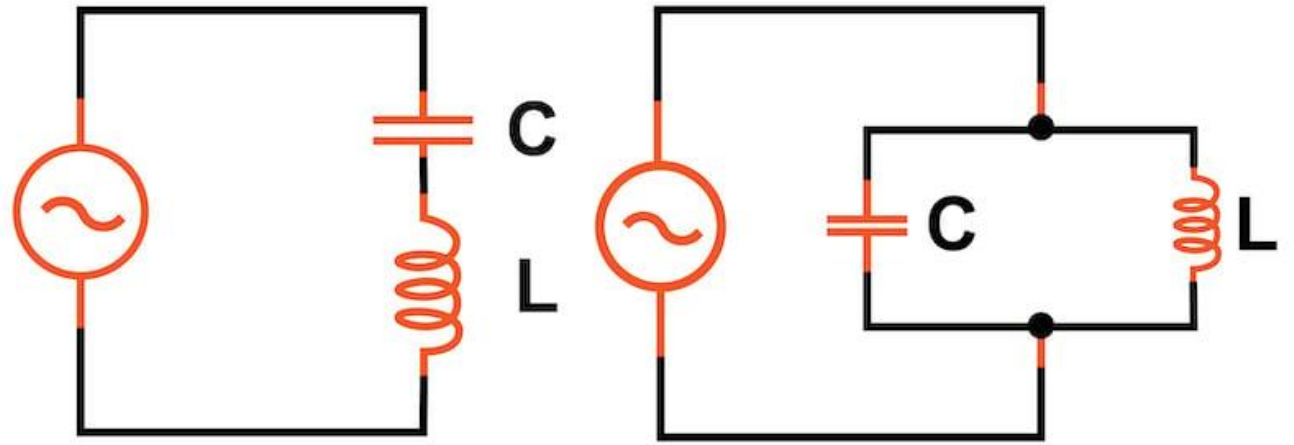
LR Series Circuit



- **Applications:**
 - Electromagnets, motor windings, choke circuits, etc.

LC Circuit

- Energy **oscillates** between the electric field of capacitor and the magnetic field of inductor.
- **Resonant Frequency:**
- Oscillations are sinusoidal.
- **Applications:**
 - Tuned circuits, oscillators, radio frequency selection, etc.



$$X_L = X_C$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

RCL Circuits

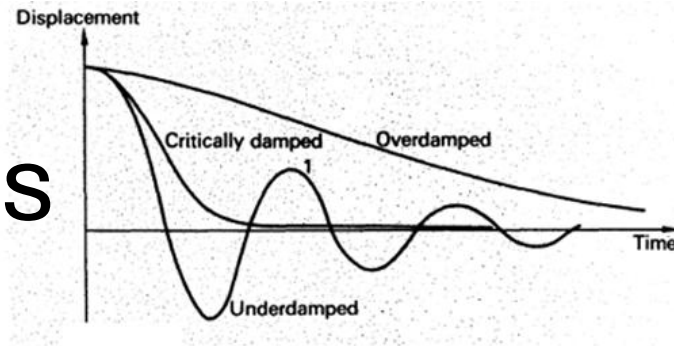
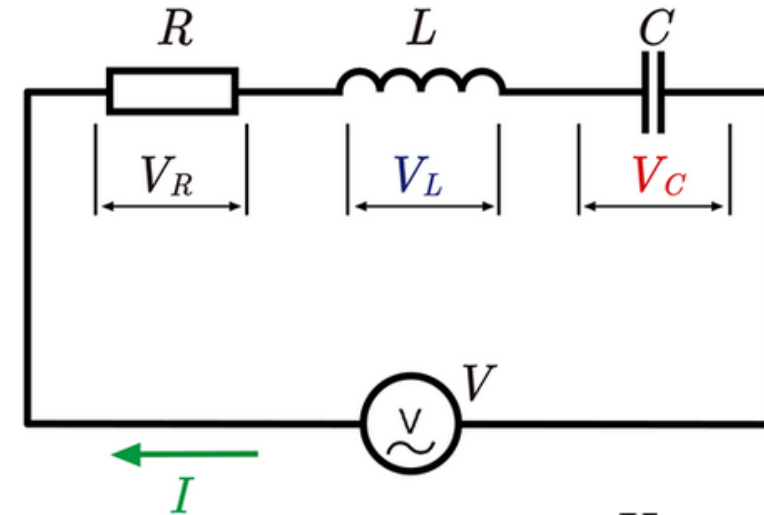


Fig.3. 4 Types of damped oscillation

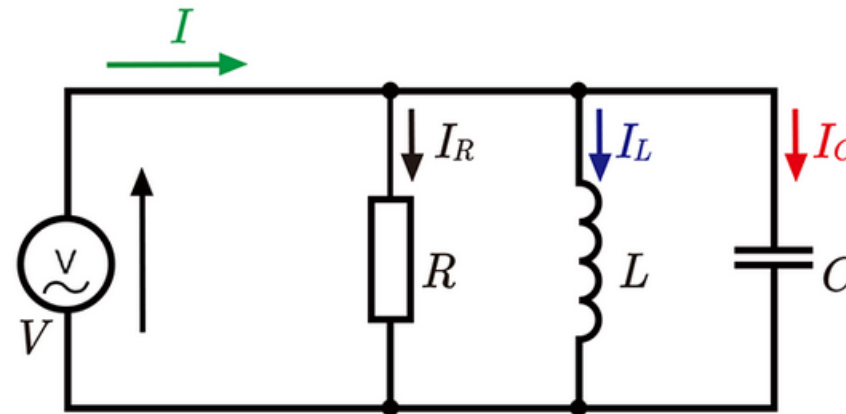
- Series or parallel
- **Damped Oscillations:**
 - Resistance causes energy loss.
 - Oscillations gradually decrease.
- **Resonance (AC Supply):**
- **Applications:**
 - Band-pass and band-stop filters, radio and communication systems, signal processing, etc.

Series RLC Circuit

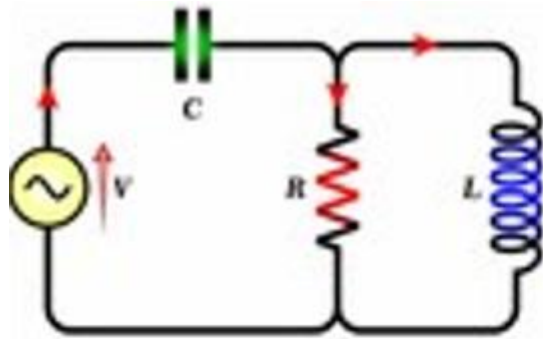


$$X_L = X_C$$

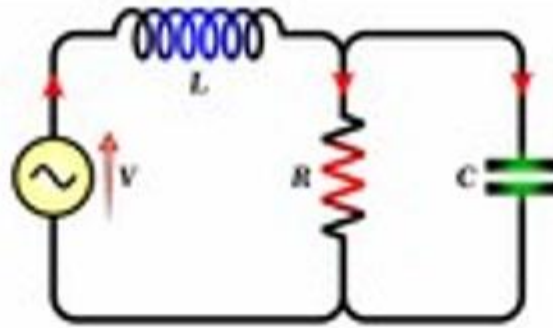
Parallel RLC Circuit



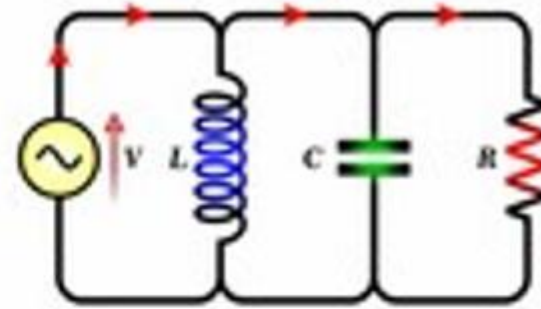
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$



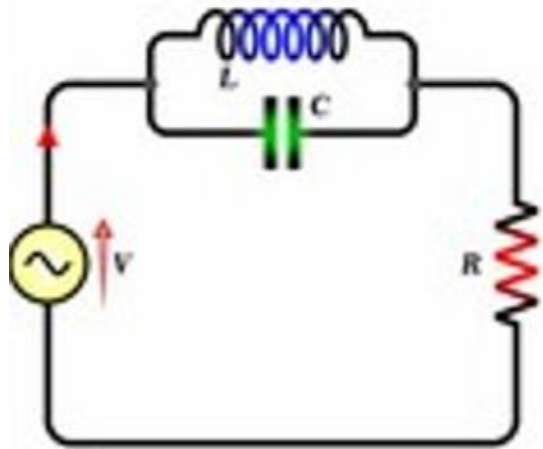
R L C circuit as a high-pass filter



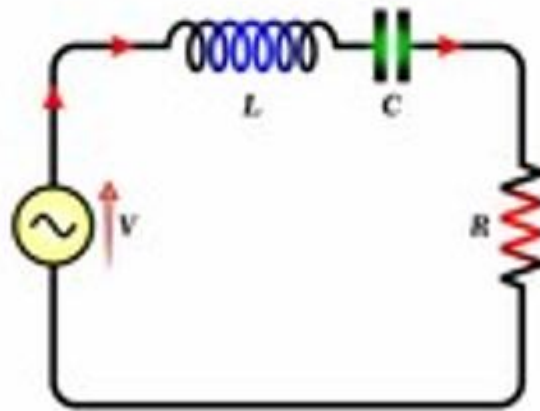
R L C circuit as a low-pass filter



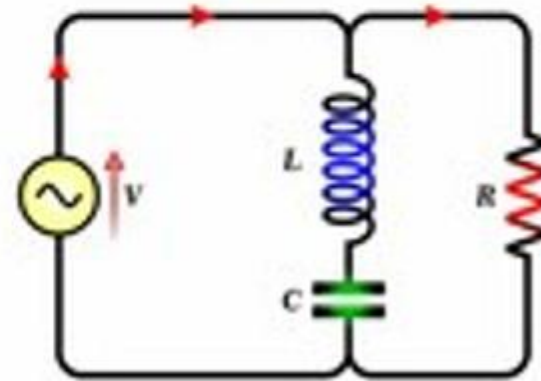
R L C circuit as a parallel band-pass filter in shunt across the line



R L C circuit as a parallel band-stop filter in series with the line



R L C circuit as a series band-pass filter in series with the line



R L C circuit as a series band-stop filter in shunt across the line

Transformers (Ideal)

- **Step-up transformer:** $N_s > N_p$
- **Step-down transformer:** $N_s < N_p$

